Exploring Quantum Computing in Particle Physics

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The theory describing a self-interacting, real, scalar quantum field $\phi(\boldsymbol{x})$, can be mapped onto a quantum computer by sampling over a discretized spatial lattice, and 'digitizing' the field such that it is constrained to a finite set of sample states. Due to the Nyquist-Shannon sampling theorem for band limited signals, with optimal tuning of parameters, the theory can be reconstructed in this way up to exponentially small error. There are two common ways to achieve digitization of the scalar field: the JLP approach, where eigenstates of the field operator ϕ are used as a digitization basis; and the HO basis approach, that uses eigenstates of a harmonic oscillator with frequency ω_ϕ. Inspired by the work of Klco and Savage, both of these approaches are investigated and compared in the context of the free scalar field and the scalar field with quartic self-interaction in 0+1 dimensions. The HO basis is identified as analogous to 1st order Hamiltonian truncation effective theory; a non-perturbative numerical method for calculating QFT observables based on truncation of the interacting Hamiltonian to a finite-dimensional effective Hilbert space spanned by eigenstates of the free Hamiltonian. Motivated by this insight, a second order improvement term is added to the HO basis digitized Hamiltonian for both free and $\lambda\phi^4$ scalar field theories, aiming to improve the scaling of digitization error with the number of qubits used for the digitization, without introducing significant noise potential through additional gate resources. Whilst decomposition of the digitized Hamiltonian into multi-Pauli gates showed that inclusion of the 2nd order term required only modest additional gate resources, exact diagonalization showed that expected percent errors in GS energy calculations on ideal quantum devices *increased* upon inclusion of the 2nd order term, yielding no evidence of an improved scaling.

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1. INTRODUCTION

Whilst the promise of Quantum Computing (QC) to transcend the limitations of classical computation and ignite a new era of computational supremacy is still a work in progress, the theoretical potential for its application is certainly evident. With potential applications in Cryptography and Security [1], Drug Discovery and Molecular Modeling [2] and Machine Learning and AI [3] to name just a few examples, QC indeed looks very promising. Despite a wealth of theoretical exploration and applications however, QC remains at research level with a practical and impactful technology currently out of reach. The limiting factor here is the development of quantum hardware, which is susceptible to errors and environmental noise, with the maintenance of quantum coherence rendering scalability difficult. Current quantum devices [4] lack error correction, have short coherence times (limiting the number of gates that can be applied), and are imprecise. There are significant efforts worldwide [5] to achieve fault tolerant, scalable devices but for the meantime, the practical application of QC will be plagued by these limitations. This interim period has been labelled the NISQ-era by John Preskill [6], referring to Noisy, Intermediate-Scale, Quantum devices, and at least for now, we are limited to theoretical exploration of the potential applications of such NISQ-era devices.

One area where the application of QC has proved prolific is the simulation of the time evolution of QFTs on quantum computers and the development of quantum algorithms to calculate S-matrix elements and scattering amplitudes [7-12]. Whilst QFT is predominantly considered a great success, it is not without its limitations: perturbative methods are limited to weakly interacting theories, and non-perturbative methods for strong coupling, such as lattice QCD, involve great computational complexity [13]. Quantum computing may provide a resolution to these limitations: in a lecture in 1981 [14], Feynman first proposed the idea that devices which utilise quantum phenomena such as superposition and entanglement would be inherently suited to the simulation of quantum systems, offering the potential for non-perturbative simulation of QFTs with exponential speedup over classical methods. Forty-Three years later, the recent surge in QC advancements and emerging availability of NISQ-era hardware is rendering this proposition increasingly tangible.

In a string of seminal papers [7-9], Jordan, Lee and Preskill (JLP) developed an algorithm for calculating scattering amplitudes in scalar QFTs based on discretization of the continuous field value at each lattice site. In order to reconstruct the theory on a system of qubits, space is discretized into a d-dimensional lattice of N^d points with spacing δx and extent $L = N\delta x$ in each spatial dimension, equipped with either periodic or twisted boundary conditions. The continuous real scalar field $\phi(\boldsymbol{x})$ is then sampled at each of the N^d discrete lattice points with its value restricted to a 'digitized' basis of 2^{n_q} discrete values represented on n_q qubits for each site. The result is a system of $N^d \times n_q$ qubits representing the theory, which can be evolved under the time evolution of the appropriate Hamiltonian operator. JLP show that such an algorithm scales exponentially faster than classical methods at high precision or strong coupling and in related work [10], extend this model to algorithms for fermionic theories. At present, there is a wealth of literature on the simulation of fermionic systems [10-12][14-17], with optimised fermion algorithms and mapping of fermion states to qubits well established. Pioneering work by Macridin, Spentzouris, Amundson and Harnik (MSAH) [18][19] focusing on electron-phonon interactions extended this framework by developing algorithms for fermion-boson in-

teractions, expanding understanding of boson-boson interactions in the process. In their work, phonons are modelled as a finite set of harmonic oscillators, and through considerations of the Nyquist-Shannon (NS) sampling theorem [20] in the context of eigenfunctions of the harmonic oscillator (HO) it is shown that, up to exponentially small error, the low energy space of a HO is isomorphic to the low energy subspace of a finite dimensional Hilbert space. From here, an improved algorithm is proposed for scalar fields which, similarly to the JLP algorithm, uses eigenstates of the field operator as a digitization basis, except now with built is NS bound saturation. It is shown that the qubit requirements for this improved algorithm scale exponentially faster than those of the JLP algorithm with $n_q \sim |\log|\log|\epsilon_{dig}|||$ rather than $n_q \sim |\log|\epsilon_{dig}||$, where ϵ_{dig} is the error due to digitization only.

It should by now be evident that the game here is to maximise precision of calculations whilst keeping resource requirements (e.g. the number of qubits, n_q) as low as possible. Digressing briefly from QC, in a recent paper [21], Cohen, Farnsworth, Houtz and Luty propose a non-perturbative numerical method for calculating QFT observables based on truncation of the interacting Hamiltonian $H = H_0 + V$ to a finite-dimensional effective Hilbert space spanned by eigenstates of the free Hamiltonian up to some cut-off energy: $\mathcal{H}_{eff} := span\left\{|i\rangle\right\}_{E_i \leq E_{max}}$ where $H_0|i\rangle = E_i|i\rangle$. An effective Hamiltonian is computed in this truncated space by matching a 'transition matrix' to the full theory, yielding corrections order by order as an expansion in powers of $1/E_{max}$ (i.e. $H_{eff} = H_0 + H_1 + H_2 + ...$) such as to retain information from the higher energy states that have been truncated away. This method, appropriately coined 'Hamiltonian Truncation Effective Theory' (HTET) appears immediately relevant to the reconstruction of scalar QFTs on quantum computers, where full theory Hamiltonians must be approximated in finite-dimensional spaces in order to be represented on digital devices. The motivation behind this paper is to investigate the implications of HTET in this context, specifically, whether it can be used to leverage improved precision in calculations without significantly increasing resource requirements.

The natural place to begin such an investigation is the digitization process by which the continuous field value is discretized and represented via a finite-dimensional basis so that it can be represented on a digital device. A paper by Klco and Savage (KS) focusing on the digitization of scalar fields [22] nicely ties together the JLP algorithm for digitization [7-9] with the insights from MSAH [18] on the significance of the NS theorem, and shows that upon 'saturation of the NS bound', digitization errors for the JLP algorithm scale as $|\log|\log|\epsilon_{dig}|||\sim n_q$ in agreement with the MSAH algorithm. Further to this, KS propose an alternative digitization basis using eigenstates of the harmonic oscillator to discretize the field rather than eigenstates of the field operator, and examine the efficacy of this approach in comparison to the approach taken by JLP. Their investigation is conducted within the context of three example systems: free scalar field theory in $0+1$ dimensions (the harmonic oscillator), and $\lambda\phi^4$ scalar field theory in both $0+1$ and $1+1$ dim.

The first part of this paper works alongside Klco and Savage's investigation, reproducing plots and deriving insights by exploring relevant papers. It aims to obtain a deeper understanding of the digitization process; namely how these different bases compare and how they can be tuned to optimise precision of calculations (minimise error due to digitization) with minimal resource requirements. This effort establishes a robust groundwork for further exploration of HTET in the second part of this paper, where it is recognised that the HO basis digitized

Hamiltonian described by KS is analogous to the effective Hamiltonian described by HTET to first order, corresponding to raw truncation of the full Hamiltonian in the effective Hilbert space. The remainder of the paper seeks to use the matching conditions described in ref [21] to derive the second order improvement terms of the effective Hamiltonian for three example systems, free scalar field theory and $\lambda\phi^4$ scalar field theory in both weakly and strongly coupled regimes in 0+1 dimensions before assessing the implications of this second order 'improvement' compared with the first order case. It is shown that although promisingly requiring minimal additional resources to implement ($\sim 28\%$ more CNOT gates in the $n_q = 4$ case), the second order improvement derived from HTET fails to improve accuracy of calculations, at least for quantum mechanics.

2. LATTICE DISCRETIZATION OF SCALAR FIELD THEORY WITH QUBITS

To set the stage for our investigation into digitization, we first briefly work through lattice scalar field theory, following a similar route to that of Klco and Savage [22] for the spatial lattice in section 2.1 (though adopting different notation and lattice definitions), before closely following work of Bauer, Freystis and Nachman (BFN) [23] on the momentum-space lattice in section 2.2. The purpose of this preliminary discussion is to contextualize digitization within the lattice discretization framework; it will turn out that the digitization process is implemented analogously at each lattice site and as such, we can simplify our analysis of digitization by considering the case of $0 + 1$ dimensions consisting of a single lattice site.

2.1. Defining Scalar Field Theory on the Spatial Lattice

We begin with the Hamiltonian density for a scalar field with quartic self-interaction:

$$\mathcal{H} = \frac{1}{2}\Pi^2 + \frac{1}{2}(\nabla\phi)^2 + \frac{1}{2}m^2\phi^2 + \frac{\lambda}{4!}\phi^4. \tag{1}$$

To reconstruct the theory on a discrete number of qubits, the continuous fields are sampled at discrete points in space. This is achieved by discretizing space with a d-dimensional lattice with spacing δx and extent $L = N\delta x$ in each spatial dimension. BFN [23] define the spatial lattice with periodic boundary conditions as[1]

$$\boldsymbol{x_r} = -\boldsymbol{x}_{max} + \boldsymbol{r}\delta x, \quad \boldsymbol{x}_{max} = (n_0\delta x,\ \ldots,\ n_0\delta x)\,, \quad n_0 = \left\lfloor \frac{N-1}{2} \right\rfloor \tag{2}$$

for $\boldsymbol{r} = (r_1, \ldots, r_d)$, $\quad r_i \in \{0, \ldots, N-1\}$, and where $\lfloor y \rfloor := \max\{m \in \mathbb{Z} \mid m \leq y\}$ denotes the floor of y. The discretized Hamiltonian is now defined from the full Hamiltonian by sampling operators at each discrete lattice site. The discretized Hamiltonian for the $(d+1)$ dim. theory on a spatial lattice defined as above takes the form:

$$H = (\delta x)^d \frac{1}{2} \sum_{\boldsymbol{x}} \left[{\Pi_{\boldsymbol{x}}}^2 - \phi_{\boldsymbol{x}} \left[\boldsymbol{\nabla}^2\phi\right]_{\boldsymbol{x}} + {m_0}^2 {\phi_{\boldsymbol{x}}}^2 + \frac{\lambda_0}{4!}{\phi_{\boldsymbol{x}}}^4 \right], \tag{3}$$

[1] Note that in this formalism, the distance between the first and last point in each spatial dimension is $(N-1)\,\delta x = L - \delta x$. The extra spacing in the lattice extent $L = N\delta x$ is in fact the jump between lattice 'wraps' to satisfy P.B.C.s.

where $\phi_{\boldsymbol{x}} \equiv \phi(\boldsymbol{x}), \Pi_{\boldsymbol{x}} \equiv \dot{\phi}(\boldsymbol{x})$ and $\boldsymbol{x} \in \{\boldsymbol{x_r}\}$. We have the equal-time commutation relation:

$$[\phi_{\boldsymbol{x}}, \Pi_{\boldsymbol{x}}] = \frac{i}{(\delta x)^d} \delta^{(d)}_{\boldsymbol{x},\boldsymbol{y}} \hat{I}. \tag{4}$$

The discretized Laplacian operator $[\boldsymbol{\nabla}^2 \phi]_{\boldsymbol{x}}$ is defined via finite-difference approximations; a d-dimensional analogue of the central difference approximation is used to approximate partial derivatives via differences in the field value at neighbouring lattice sites. In this way, (see Appendix A for a detailed derivation) the discretized Laplacian operator can be defined:

$$\left[\boldsymbol{\nabla}^2 \phi\right]_{\boldsymbol{x}} = \sum_{i=1}^{d} \frac{1}{(\delta x)^2} [-2\phi_{\boldsymbol{x}} + \phi_{\boldsymbol{x}+\delta x \hat{\boldsymbol{\mu}}_i} + \phi_{\boldsymbol{x}-\delta x \hat{\boldsymbol{\mu}}_i}], \tag{5}$$

where the $\hat{\boldsymbol{\mu}}$ are unit vectors in each spatial dimension. At this point, Klco and Savage redefine the fields, Hamiltonian and mass: $\hat{\phi} = (\delta x)^{\frac{(d-1)}{2}} \phi$, $\hat{\Pi} = (\delta x)^{\frac{(d+1)}{2}} \Pi$, $\hat{H} = \delta x H$, $\hat{m}_0 = \delta x m_0$, $\hat{\lambda}_0 = (\delta x)^{3-d} \lambda_0$ [22] to obtain the dimensionless Hamiltonian

$$\hat{H} = \frac{1}{2} \sum_{\boldsymbol{x}} \left[\hat{\Pi}^2_{\boldsymbol{x}} - \hat{\phi}_{\boldsymbol{x}} \left[\hat{\boldsymbol{\nabla}}^2 \hat{\phi} \right]_{\boldsymbol{x}} + {m_0}^2 \hat{\phi}^2_{\boldsymbol{x}} + \frac{\hat{\lambda}_0}{4!} \hat{\phi}^4_{\boldsymbol{x}} \right], \tag{6}$$

and corresponding equal-time commutation relation

$$\left[\hat{\phi}_{\boldsymbol{x}}, \hat{\Pi}_{\boldsymbol{x}}\right] = i \delta^{(d)}_{\boldsymbol{x},\boldsymbol{y}} \hat{I}. \tag{7}$$

2.2. The Free Hamiltonian on the Momentum Space Lattice

The spatial discretization equipped with either periodic or twisted boundary conditions[2] defines a corresponding lattice discretization in momentum space; we have that momentum space is quantized in each dimension with spacing $\delta k = \frac{2\pi}{N \delta x}$, $\left(k_{min} = \frac{2\pi}{\lambda_{max}}\right)$, and lattice extent $L_k = 2k_{max}$, $k_{max} = \frac{2\pi}{\lambda_{min}} = \frac{\pi}{\delta x}$, defined by the first Brillouin zone. BFN [23] define the momentum space lattice as[3]

$$\boldsymbol{k_s} = -\boldsymbol{k}_{max} + (\boldsymbol{s} - \boldsymbol{\Delta})\, \delta k, \quad \boldsymbol{k}_{max} = (n_0 \delta k,\ \ldots,\ n_0 \delta k), \quad n_0 = \left\lfloor \frac{N-1}{2} \right\rfloor \tag{8}$$

where $\boldsymbol{s} = (s_1, \ldots, s_d), \quad s_i \in \{0, \ldots, N-1\}, \quad$ and now we also have $\boldsymbol{\Delta} = (0, \ldots, 0)$ for periodic boundary conditions and $\boldsymbol{\Delta} = (\frac{1}{2}, \ldots, \frac{1}{2})$ for twisted boundary conditions.[4]

Now that we have defined both the spatial and momentum-space lattices, we define the discrete Fourier transform to transition between them. Adopting the convention used by BFN [23], the discrete Fourier transform (DFT) is

$$\phi_{\boldsymbol{x}} = \frac{1}{(2\pi)^d} \sum_{\boldsymbol{k}} e^{-i\boldsymbol{k}\cdot\boldsymbol{x}} \phi_{\boldsymbol{k}}, \quad \phi_{\boldsymbol{k}} = \sum_{\boldsymbol{x}} e^{i\boldsymbol{k}\cdot\boldsymbol{x}} \phi_{\boldsymbol{x}}, \tag{9}$$

[2] Periodic boundary conditions with a relative minus sign for each 'wrap' around the lattice.

[3] Note that the $\boldsymbol{k}_{max}$ in this formalism is not actually the Brillouin zone boundary however it is straightforward to see that the $\boldsymbol{k}$ values lie on or within the boundaries, with the extent in each dimension (including the extra spacing between lattice 'wraps') being $\frac{2\pi}{\delta x}$ as required.

[4] To see why $\boldsymbol{\Delta} = (\frac{1}{2}, \ldots, \frac{1}{2})$ corresponds to T.B.C.s, see Appendix B. In short, by using (9) to go to momentum space and back we can write $\phi_{\boldsymbol{x}+L\hat{\boldsymbol{\mu}}_{\boldsymbol{i}}} = e^{i(2\pi\Delta_i)} \phi_{\boldsymbol{x}}$ where $L = N\delta x$ is the spatial lattice extent. Thus half-integer values of Δ_i give rise to a relative minus sign in the boundary condition, equivalent to T.B.C.s.

with the completeness relation

$$\frac{1}{(2\pi)^d}\sum_{\boldsymbol{k}} e^{-i\boldsymbol{k}\cdot(\boldsymbol{x}-\boldsymbol{x}')}\phi_{\boldsymbol{p}} = \delta_{\boldsymbol{x}\boldsymbol{x}'}, \quad \sum_{\boldsymbol{x}} e^{i\boldsymbol{x}\cdot(\boldsymbol{k}-\boldsymbol{k}')}\phi_{\boldsymbol{x}} = \delta_{\boldsymbol{k}\boldsymbol{k}'}. \tag{10}$$

We can now express our discretized scalar field theory on the momentum-space lattice. We do this for the free theory. Starting with the discretized Hamiltonian for the free scalar field theory defined on the spatial lattice

$$\hat{H} = \frac{1}{2}\sum_{\boldsymbol{x}}\left[\hat{\Pi}_{\boldsymbol{x}}^2 - \hat{\phi}_{\boldsymbol{x}}\left[\hat{\boldsymbol{\nabla}}^2\hat{\phi}\right]_{\boldsymbol{x}} + {m_0}^2\hat{\phi}_{\boldsymbol{x}}^2\right], \tag{11}$$

(9) and (10) are used to perform a discrete Fourier transform to the momentum-space lattice. After a somewhat lengthy procedure (the full calculation can be found in Appendix C), one arrives at the momentum-space representation

$$\hat{H} = \frac{1}{2}\cdot\frac{1}{(2\pi)^d}\sum_{\boldsymbol{k}}\left[\hat{\Pi}_{-\boldsymbol{k}}\hat{\Pi}_{\boldsymbol{k}} + \omega_{\boldsymbol{k}}^2\hat{\phi}_{-\boldsymbol{k}}\hat{\phi}_{\boldsymbol{k}}\right], \tag{12}$$

where the frequency

$$\omega_{\boldsymbol{k}}^2 = m_0^2 + \frac{4}{(\delta x)^2}\sum_{i=1}^{d}\sin^2\left(\frac{k_i\delta x}{2}\right). \tag{13}$$

and we deduce that the discretized free theory reduces to a collection of uncoupled harmonic oscillators on the momentum-space lattice .

2.3. Digitization and Qubit Representation

To represent the Hilbert space of the free scalar field theory on a system of finite qubits, further adjustments must be made beyond lattice discretization; the missing piece is the process of 'digitization', which in the present context, is best seen from two angles:

- From the perspective of position-space, we have so far discretized spatial coordinates to a lattice consisting of N^d sample sites. However, the space of potential field values each lattice site can take is still infinite and continuous. To represent this field space on a system of n_q qubits it must be digitized, i.e. truncated and discretized such that the value of the scalar field at each site is limited to 2^{n_q} discrete sample values[5].
- Alternatively, from the perspective of the momentum-space lattice, we have an uncoupled harmonic oscillator at each lattice site. The spectrum of each harmonic oscillator can be represented on n_q qubits by restricting to the lowest 2^{n_q} states and identifying these with the 2^{n_q} basis states spanning our qubit space.

The details of these processes are discussed in more detail in Sections 3.2-3.3 The point here is that digitization is implemented consistently across every lattice site: the state $|a\rangle_i$ at site i

[5] The tensor product space of n_q qubits is spanned by 2^{n_q} basis states, e.g. for $n_q = 3$ we have $|000\rangle, |001\rangle, \ldots, |111\rangle$, (in the computational basis) where $|ijk\rangle := |i\rangle \otimes |j\rangle \otimes |k\rangle$

is represented on a system of n_q qubits s.t. $|a\rangle_i = \left|q_{n_q-1}\dots q_1 q_0\right\rangle_i$. The full Hilbert space is then represented by the tensor product of all of the lattice sites, $|\Psi\rangle = \otimes_{i=0}^{N^d} |a\rangle_i$, corresponding to a system of ${n_q}^{N^d}$ qubits. With the above discussion in mind, it is clear that the digitization process is no different whether we are considering just one lattice site or many, as it is implemented analogously at each site. Motivated by this observation, we can simplify our analysis of digitization without loss of generality by focusing in on the $0+1$ dimensional case, i.e. quantum mechanics, which consists of a single lattice site.

3. DIGITIZATION- THE HARMONIC OSCILLATOR

For the remainder of the paper, discussions are mostly confined to the case of 0+1 dimensions, however results generalise straightforwardly to higher dimensions. Following closely in the footsteps of Klco and Savage [22] we begin with the free scalar field theory, which in 0+1 dimensions amounts to a single harmonic oscillator at a single lattice site.

3.1. The Quantum Harmonic Oscillator

In 0+1 dimensions, our discretized Hamiltonian for the free scalar field theory simplifies to

$$\hat{H} = \frac{1}{2}\hat{\Pi}^2 + \frac{1}{2}{m_0}^2\hat{\phi}^2, \tag{14}$$

where $\hat{\phi}$ and $\hat{\Pi}$ are now simply the position and momentum operators of quantum mechanics. Following Klco and Savage [22] we make a further field and Hamiltonian redefinition, $\hat{\phi} = \frac{1}{\sqrt{\hat{m}_0}}\bar{\phi}$, $\hat{\Pi} = \sqrt{\hat{m}_0}\bar{\Pi}$, $\hat{H} = \sqrt{\hat{m}_0}\bar{H}$, effectively setting $\bar{m}_0 = 1$ to obtain the Hamiltonian

$$\bar{H} = \frac{1}{2}\bar{\Pi}^2 + \frac{1}{2}\bar{\phi}^2 \quad \text{where} \quad \left[\bar{\phi}, \bar{\Pi}\right] = i\hat{I}. \tag{15}$$

at equal times. This system is simply the free harmonic oscillator with $\omega = \bar{m}_0 = 1$. This is the same system that was studied by MSAH [18] in their discussion of the Nyquist-Shannon Sampling theorem, albeit with different notation: MSAH use X and P to denote the position and momentum operators as this is simply quantum mechanics after all, however we continue with the field notation for consistency when generalising to field theory.

Our Hamiltonian has an infinite tower of eigenstates

$$|n\rangle = \int \psi_n(\phi)\,|\phi\rangle\,d\phi = \int \widetilde{\psi}_n(k)\,|k\rangle\,dk \tag{16}$$

s.t. $\bar{H}\,|n\rangle = E_n\,|n\rangle$ with eigenvalues $E_n = n + \frac{1}{2}$. The

$$\psi_n(\phi) = \frac{1}{\pi^{\frac{1}{4}}\sqrt{2^n n!}} e^{-\frac{\phi^2}{2}} \boldsymbol{H}_n(\phi) \tag{17}$$

are the Hermite-Gauss (HG) functions of order n and the $|\phi\rangle$ and $|k\rangle$ correspond to the field space and conjugate-momentum space coordinate bases resp. The $\widetilde{\psi}_n(k)$ denote the HG functions in momentum space which, as noted by MSAH take the form

$$\widetilde{\psi}_n(k) = \frac{(-i)^n}{\pi^{\frac{1}{4}}\sqrt{2^n n!}} e^{-\frac{k^2}{2}} \boldsymbol{H}_n(k), \tag{18}$$

since the $\psi_n(\phi)$ are also eigenfunctions of the Fourier transform operator [18]:

$$[\mathcal{F}(\psi_n)](k) \equiv \widetilde{\psi}_n(k) = (-i)^n \psi_n(k). \tag{19}$$

Now that we have established the foundational properties of the harmonic oscillator, we can proceed to define the formalism for its digitization.

3.2. The JLP Basis

In the algorithm proposed by Jordan, Lee and Preskill [7-9], as well as the improved algorithm proposed by MSAH [18], digitization is achieved by using eigenstates of the field operator to discretize field space. To reconstruct the field operator on n_q qubits, it must be constrained to take 2^{n_q} discrete values. This is achieved by discretizing the continuous field space in a way analogous to the 1-dimensional case of the spatial lattice; we truncate the field space and constrain it to a set of discrete values within this truncated range. The digitized field operator is then defined through its action on 2^{n_q} eigenstates with eigenvalues corresponding to the 2^{n_q} sample values. The action of this digitized field operator on these eigenstates can then be reconstructed on a quantum circuit by identifying the 2^{n_q} eigenstates with the 2^{n_q} basis states spanning the tensor product space of our n_q qubits, before decomposing the action of the field operator on these states into tensor products of unitary qubit gates. To make this more concrete, we now introduce the mathematical formalism for the JLP basis approach to digitization in the context of the harmonic oscillator, as presented by Klco and Savage [22].

i. Basis Definition

The digitized field operator $\tilde{\phi}$ introduced by JLP, as described by Klco and Savage [22], is defined by its action on $n_s = 2^{n_q}$ basis states $\left|\tilde{\phi}^\alpha\right\rangle$ for an n_q qubit system

$$\tilde{\phi}\left|\tilde{\phi}^\alpha\right\rangle = \tilde{\phi}^\alpha\left|\tilde{\phi}^\alpha\right\rangle \tag{20}$$

where

$$\tilde{\phi}^\alpha = -\bar{\phi}_{max} + \alpha\delta_{\tilde{\phi}} \quad \text{for} \quad \alpha \in \{0, 1, 2, \ldots, n_s - 1\} \tag{21}$$

are the $n_s = 2^{n_q}$ sample values of the digitized field space. $\bar{\phi}_{max}$ is the truncation value which we are free to tune, and

$$\delta_{\tilde{\phi}} = \frac{2\bar{\phi}_{max}}{n_s - 1} \tag{22}$$

is the sample spacing of our digitization. Note that this definition of field space digitization is entirely analogous to the 1-dimensional case of the spatial lattice defined previously (2) with $N \to n_s$ odd and $\delta x \to \delta_{\tilde{\phi}}$.[6] The result is a set of n_s evenly spaced sample values from $-\bar{\phi}_{max}$ to $+\bar{\phi}_{max}$, a visual representation of which can be seen in Fig.1, for the case of $n_q = 3$ qubits:

[6] Note that in this analogy, the lattice extent $L = 2\tilde{\phi}_{max} + \delta_{\tilde{\phi}}$ as it includes the extra spacing between lattice wraps when P.B.C.s are imposed.

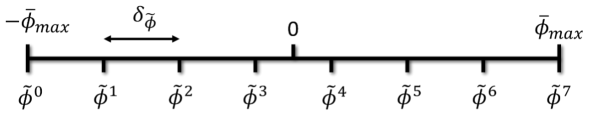


FIG. 1: The JLP digitized field space for $n_q = 3$ qubits is composed of $2^{n_q} = 8$ sample values $\tilde{\phi}^\alpha$ distributed evenly across the truncated interval $\left[-\tilde{\phi}_{max}, \tilde{\phi}_{max}\right]$ with spacing $\delta_{\tilde{\phi}}$ such that they are symmetrically centered around zero.

In analogy with lattice discretization, under the imposition of periodic or twisted boundary conditions, a corresponding digitization can be defined for the conjugate-momentum space:

$$k_{\tilde{\phi}}^{\beta} = -k_{\tilde{\phi}}^{max} + (\beta - \Delta)\,\delta k_{\tilde{\phi}} \quad \text{for} \quad \beta \in \{1, 2, \ldots, n_s\} \tag{23}$$

where

$$k_{\tilde{\phi}}^{max} = \frac{\pi}{\delta_{\tilde{\phi}}} \quad \left(k_{max} = \frac{2\pi}{\lambda_{min}}\right), \qquad \delta k_{\tilde{\phi}} = \frac{2\pi}{n_s \delta_{\tilde{\phi}}} = \frac{2k_{\tilde{\phi}}^{max}}{n_s} \quad \left(k_{min} = \frac{2\pi}{\lambda_{max}}\right) \tag{24}$$

are defined by the first Brillouin zone and $\Delta = 0,\ \frac{1}{2}$ for periodic and twisted boundary conditions respectively.[7] A visual representation of this definition is shown in FIG.2 for the case of $n_q = 3$, where the upper scale represents the first Brillouin zone and the lower scales represent the basis convention under both periodic and twisted B.C.s. Note that in contrast to the momentum-space lattice defined previously, our digitization basis here is centered to the right of the zero mode rather than to the left (for P.B.C.s);[8] this is simply convention[9] and the reason for this choice is so that for T.B.C.s, our basis states are symmetrically distributed *within* the edges of the first Brillouin zone, between values $\pm\frac{\pi}{\delta_{\tilde{\phi}}}\frac{n_s-1}{n_s}$. This is for later convenience when saturating the Nyquist-Shannon bound (see Section 3.2.*iv.*).

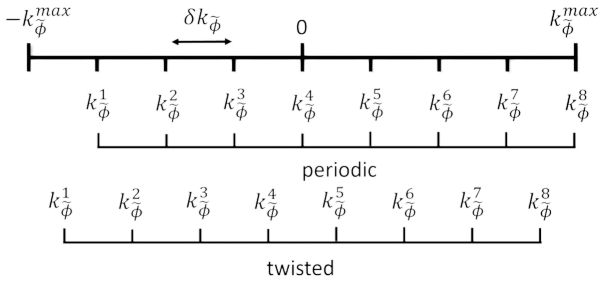


FIG. 2: A visual representation of the JLP digitization of conjugate-momentum space with $n_q = 3$ qubits. The upper scale represents the first Brillouin zone whilst the middle and lower scales represent the sample modes composing the digitization basis for P.B.C.s and T.B.C.s respectively.

ii. The Conjugate-Momentum Operator

The digitized field operator is defined trivially via the field-space digitization. We now discuss how to treat the conjugate-momentum operator acting in our digitized regime, closely following the work of Klco and Savage [22]. The standard approach is to identify the conjugate-momentum operator with a derivative in field space, $\bar{\Pi} = -i\frac{\partial}{\partial\tilde{\phi}}$. $\bar{\Pi}^2 = -\frac{\partial^2}{\partial\tilde{\phi}^2}$ can then be

[7] This result can be derived analogously to the 1-dimensional case of Appendix B.

[8] As a consequence, in contrast to the earlier lattice example, the $k_{\tilde{\phi}}^{max}$ here *does* correspond to the edge of the first Brillouin zone.

[9] we are free to do this since the Nyquist-Shannon Sampling theorem [20] (3.2.*iv.*) places no restriction on the momentum modes we use to discretize the conjugate-momentum space.

approximated by finite-difference approximations analogously to the 1-dimensional case of the discretized Laplacian operator (Appendix A) such that we obtain the finite-difference operator:

$$\tilde{\Pi}\left|\tilde{\phi}^{\alpha}\right\rangle = \frac{1}{\left(\delta_{\tilde{\phi}}\right)^{2}}\left[2\left|\tilde{\phi}^{\alpha}\right\rangle - \left|\tilde{\phi}^{\alpha+1}\right\rangle - \left|\tilde{\phi}^{\alpha-1}\right\rangle\right] \tag{25}$$

where the $\left|\tilde{\phi}^{\alpha}\right\rangle$ are the n_s basis states spanning our digitized field space. We hence have [22]

$$\left\langle\tilde{\phi}^{\alpha'}\right|\tilde{\Pi}\left|\tilde{\phi}^{\alpha}\right\rangle = \begin{bmatrix} 2 & -1 & 0 & 0 & \dots & -1 \\ -1 & 2 & -1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & -1 & 2 & -1 \\ -1 & 0 & 0 & \dots & -1 & 2 \end{bmatrix} \tag{26}$$

for P.B.C.s, with T.B.C.s resulting in $+1$s in the off-diagonal corners (due to the relative minus sign in the boundary condition). KS [22] state that by performing a Fourier transform to conjugate-momentum space, it can be shown that (Appendix D) this finite-difference operator is diagonal when acting in our digitized conjugate-momentum basis, with eigenvalues

$$\hat{k}_{\tilde{\phi}}^{2} = \frac{4}{\delta_{\tilde{\phi}}}\sin^{2}\left(\frac{k_{\tilde{\phi}}\delta_{\tilde{\phi}}}{2}\right). \tag{27}$$

We now have the digitized field operator, $\tilde{\phi}$, which is diagonal in the digitized field-space basis, and the momentum-conjugate of the finite-difference operator, $\tilde{\Pi}$, which is similarly diagonal in the digitized conjugate-momentum space basis. Consequently, the most efficient way to evaluate the action of our digitized Hamiltonian on a system of qubits is to first directly evaluate the diagonal $\tilde{\phi}^2$ operator in the digitized field-space basis before performing a quantum Fourier transform of the qubit register, transforming it to the digitized conjugate-momentum basis upon which the finite-difference operator can be directly evaluated in its diagonal form (see Section 3.2.*iii*). However, KS explain that whilst this formulation yields exponentially convergent quantities, due to the difference between $\hat{k}_{\tilde{\phi}}$ and $k_{\tilde{\phi}}$ in (27), these convergent quantities (specifically eigenenergies) differ from the true HO quantities by even powers of $\delta_{\tilde{\phi}}$. KS consequently propose a method [22] to systematically improve the Hamiltonian in powers of ${\delta_{\tilde{\phi}}}^2$, but ultimately conclude that the resource requirements for such an effort are excessive and unreasonable. An alternative, more convenient approach is proposed for handling the conjugate-momentum operator, one which circumvents the issue entirely. It turns out that if instead of using the momentum-conjugate of the finite-difference operator, we define a similar diagonal operator, whose eigenvalues now correspond to the $k_{\tilde{\phi}}$ rather than the $\hat{k}_{\tilde{\phi}}$ when acting in our digitized conjugate-momentum basis, our Hamiltonian yields eigenenergies which converge exponentially[10] to those of the true, undigitized theory. It turns out that such an operator corresponds to the use of the exact conjugate-momentum operator $\bar{\Pi}$.

[10] The relative error due to digitization, ϵ_{dig}, decreases exponentially as the number of qubits, n_q, used to digitize the theory increases: $n_q \sim |\log|\epsilon_{dig}||$.

iii. Qubit Representation of Operators

Now that we have defined how the operators composing our Hamiltonian act in our digitized regime, we digress briefly from discussion of efficacy to more concretely define how the action of the Hamiltonian is manifested on the qubit register. As discussed briefly in section 2.3, the digitized field operator is realised by first identifying the $\left|\tilde{\phi}^{\alpha}\right\rangle$ which span the digitized field space, with the computational basis states, $|\alpha\rangle$, which span the tensor product space of the qubit register, where the integer α is represented via its binary representation across the n_q qubits.[11] The action of the digitized field operator on these basis states is then decomposed into tensor products of unitary qubit operators which act on the qubit register to evolve the state. We have that the digitized field operator acts diagonally on the computational basis as:

$$\tilde{\phi}\,|\alpha\rangle = \tilde{\phi}^{\alpha}\,|\alpha\rangle \tag{28}$$

i.e. for $n_q = 3$ qubits s.t. $n_s = 2^{n_q} = 8$, the matrix representing $\tilde{\phi}$ is

$$\tilde{\phi} \equiv diag\left[\tilde{\phi}^0, \tilde{\phi}^1, \ldots, \tilde{\phi}^7\right] \tag{29}$$

and trivially,

$$\tilde{\phi}^2 \equiv diag\left[\left(\tilde{\phi}^0\right)^2, \left(\tilde{\phi}^1\right)^2, \ldots, \left(\tilde{\phi}^7\right)^2\right] \tag{30}$$

Repeating for the action of the exact $\bar{\Pi}^2$ operator in the conjugate-momentum basis we obtain[12]

$$\bar{\Pi}^2 \equiv diag\left[\left(k^1_{\tilde{\phi}}\right)^2, \left(k^2_{\tilde{\phi}}\right)^2, \ldots, \left(k^8_{\tilde{\phi}}\right)^2\right]. \tag{31}$$

There are 4 unitary qubit operators that can be applied to each individual qubit: the three Pauli gates $\hat{\sigma}^x, \hat{\sigma}^y, \hat{\sigma}^z$, and the identity $\hat{\mathbb{I}}_2$. Building upon the methodology outlined by KS, one can define the general tensor product of n_q such operators

$$T^{i_1,i_2,\ldots,i_{n_q}} = \bar{\sigma}^{i_1} \otimes \bar{\sigma}^{i_2} \otimes \cdots \otimes \bar{\sigma}^{i_{n_q}} \quad \text{where} \quad \bar{\sigma} = \left(\hat{\sigma}^x, \hat{\sigma}^y, \hat{\sigma}^z, \hat{\mathbb{I}}_2\right) \tag{32}$$

The target operator $\mathcal{O}$ can be expressed as a sum over these general tensor products

$$\mathcal{O} = \sum_{i_1,i_2,\ldots,i_{n_q}} C_{i_1,i_2,\ldots,i_{n_q}} T^{i_1,i_2,\ldots,i_{n_q}} \tag{33}$$

for some coefficients $C_{i_1,i_2,\ldots,i_{n_q}}$. The action of $\mathcal{O}$ can hence be decomposed into qubit operators by projecting against the $T^{i_1,i_2,\ldots,i_{n_q}}$:

$$\begin{aligned} Tr\left[\mathcal{O}T^{i'_1,i'_2,\ldots,i'_{n_q}}\right] &= Tr\left[\sum_{i_1,i_2,\ldots,i_{n_q}} C_{i_1,i_2,\ldots,i_{n_q}} T^{i_1,i_2,\ldots,i_{n_q}} T^{i'_1,i'_2,\ldots,i'_{n_q}}\right] \\ &= \sum_{i_1,i_2,\ldots,i_{n_q}} C_{i_1,i_2,\ldots,i_{n_q}} Tr\left[T^{i_1,i_2,\ldots,i_{n_q}} T^{i'_1,i'_2,\ldots,i'_{n_q}}\right] \\ &= \sum_{i_1,i_2,\ldots,i_{n_q}} C_{i_1,i_2,\ldots,i_{n_q}} 2^{n_q} \delta^{i_1 i'_1} \delta^{i_2 i'_2} \ldots \delta^{i_{n_q} i'_{n_q}} \\ &= 2^{n_q} C_{i'_1,i'_2,\ldots,i'_{n_q}} \end{aligned} \tag{34}$$

[11] For example with $n_q = 3$ qubits we have $|0\rangle = |000\rangle\,,\ \ |1\rangle = |001\rangle\,,\ \ldots,\ \ |n_s - 1\rangle = |111\rangle$.

[12] Recall that by convention, $\beta \in \{1, 2, \ldots, n_s\}$ whereas $\alpha \in \{0, 1, \ldots, n_s - 1\}$

where in the 3rd equality we used the identity:

$$Tr\left[T^{i_1,i_2,\ldots,i_{n_q}}T^{i'_1,i'_2,\ldots,i'_{n_q}}\right] = 2^{n_q}\delta^{i_1 i'_1}\delta^{i_2 i'_2}\ldots\delta^{i_{n_q} i'_{n_q}} \tag{35}$$

In practice, this process is carried out coefficient by coefficient by expressing the operator as a projector, e.g. for $\mathcal{O} \equiv \tilde{\phi}^2$, $n_q = 2$

$$\tilde{\phi}^2 \equiv \left(\tilde{\phi}^0\right)^2 |00\rangle\langle 00| + \left(\tilde{\phi}^1\right)^2 |01\rangle\langle 01| + \left(\tilde{\phi}^2\right)^2 |10\rangle\langle 10| + \left(\tilde{\phi}^3\right)^2 |11\rangle\langle 11| \tag{36}$$

For a more explicit example of this procedure, see Appendix D. After conducting such a decomposition for our $\tilde{\phi}^2$ and $\bar{\Pi}^2$ operators, one finds that they take exactly the same form up to an overall factor, when $\tilde{\phi}^2$ acts in the field-space basis and $\bar{\Pi}^2$ acts in the conjugate-momentum basis. KS label this shared operator structure $\mathcal{O}_0^{(n_q)}$ such that for e.g. $n_q = 3$ qubits, we have $\tilde{\phi}^2 = \frac{4}{49}\bar{\phi}^2_{max}\mathcal{O}_0^{(n_q=3)}$, $\bar{\Pi}^2 = \frac{49\pi^2}{64\bar{\phi}^2_{max}}\mathcal{O}_0^{(n_q=3)}$. See the appendices of [22] for a more detailed discussion on explicit circuit constructions of these decomposed operators. As mentioned above, we first act with the decomposed $\tilde{\phi}^2$ operator on our circuit before transforming our qubit space to conjugate-momentum space and applying the decomposed $\bar{\Pi}^2$ operator. This process is clearly demonstrated in FIG.3 taken from [22], which shows the circuit required to perform Trotterized time evolution [24] of the JLP digitized harmonic oscillator Hamiltonian $\tilde{H}$, in the case of $n_q = 3$ qubits. $\Phi_3(\theta) = e^{-i\theta\mathcal{O}_{03}^{(n_q=3)}}$ is the circuit implementing the exponentiated action of $\mathcal{O}_{03}^{(n_q=3)}$,[13] for arbitrary angle θ such that the first panel corresponds to $\tilde{\phi}^2$ and the second to $\bar{\Pi}^2$. QFT_{sym} represents the symmetric quantum Fourier transform

$$|x\rangle = \frac{1}{2^{n/2}}\sum_{k=-\frac{2^n-1}{2}}^{+\frac{2^n-1}{2}} e^{\frac{2\pi i x\cdot k}{2^n}}|k\rangle\,, \tag{37}$$

that is used to transform the qubit space from digitized field space to digitized conjugate-momentum space. It effectively performs a centred, discrete Fourier transform of our qubit basis, differing from the standard QuFoTr in the fact that the resulting momentum basis is centered around the zero mode. Again, see [22] for explicit circuit construction of the symQuFoTr. The crossing of the lines represents a reversal of convention for qubit ordering, which occurs to account for the reversal of qubit order resulting from the QuFoTr without the need to utilize SWAP gates.

iv. The Nyquist-Shannon Sampling Theorem and Optimal Field-Space Truncation

It was the work of MSAH [18][19] that first emphasised the significance of the Nyquist-Shannon (NS) sampling theorem in the context of digitization of the harmonic oscillator. Through its consideration, MSAH devised a digitization algorithm that scales exponentially faster than the JLP algorithm in the number of qubits n_q. Motivated by this result, KS [22] investigated the implications of the NS sampling theorem in the context of the JLP formulation of digitization. Specifically, they investigated how the field-space truncation value, $\bar{\phi}_{max}$,

[13] $\mathcal{O}_0^{(n_q=3)}$ without the identity $\hat{\mathbb{I}}_8$.

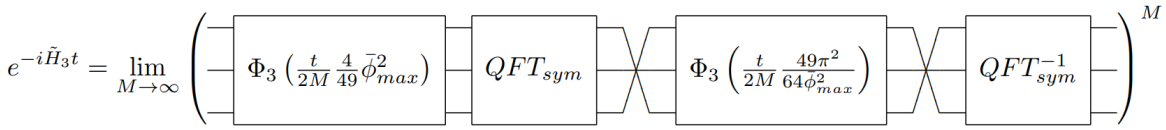


FIG. 3: Taken from [22], the circuit structure for Trotterized time evolution [24] of the JLP digitized HO Hamiltonian for a system of three qubits. The first and third panels represent the circuit implementation of the exponentiated $\theta\mathcal{O}_{03}^{(n_q=3)}$ with the argument in the first panel corresponding to $\tilde{\phi}^2$ and the argument of the second corresponding to $\bar{\Pi}^2$ acting in conjugate-momentum space. The second and fourth panels represent the symmetric quantum Fourier transform and its inverse respectively, where the line crossing indicates a reversal of qubit ordering. The diagram effectively shows how the symmetric QuFoTr is utilised to apply the $\bar{\Pi}^2$ operator in momentum space with the exact same form as $\tilde{\phi}^2$ in field space. This shared operator structure is a consequence of the symmetry of our conjugate-momentum basis and hence, if the standard QuFoTr were used instead, the resulting circuit would be significantly more complex.

affects the accuracy to which eigenstates of the HO can be reconstructed on an ideal quantum computer, showing via a plot of percent error in eigenenergies for varied $\bar{\phi}_{max}$ that for a given number of qubits, there is an optimal $\bar{\phi}_{max}$ value which reproduces the exponential improvement attained by MSAH. They claim that this optimal truncation value corresponds to "saturation of the NS bound" however give no analytical derivation or explanation of what this means. Elsewhere, in a paper by Bauer, Freytsis and Nachman on the use of effective field theories for simulating QFTs on quantum computers [23], an analytical formula is proposed for an optimal $\bar{\phi}_{max}$ value in the context of the JLP digitized free scalar field theory in general d+1 dimensions. The remainder of this section seeks to derive the 0+1 dimensional version of this formula through considerations of the NS sampling theorem in the context of the Hermite-Gauss functions (eigenfunctions of the HO) and by reproducing the plot in [22], demonstrate that this formula agrees with the optimal $\bar{\phi}_{max}$ values attained by KS. The first step in this pursuit is to define what the NS sampling theorem means in our present context.

Originating in the field of digital signal processing, the Nyquist-Shannon sampling theorem applied in this context states that given a real function $f(\phi)$ with support only in the interval $[-\phi_{max}, +\phi_{max}]$ in field space and $[-k_{max}, +k_{max}]$ in conjugate-momentum space, the function can be reconstructed by discrete sampling up to *no loss of information* provided the sample spacing in field-space, δ_ϕ, satisfies $\delta_\phi < \frac{\pi}{k_{max}}$. In the case of the digitized HO, the JLP digitization of field space corresponds to sampling the HO eigenfunctions, or HG functions $\psi_n(\phi)$, at n_s discrete sample values $\tilde{\phi}^\alpha$ which lie in the range $\left[-\bar{\phi}_{max}, +\bar{\phi}_{max}\right]$. A rough sketch of this sampling process can be seen in FIG.5 for the simple case where $\psi_0(\phi)$ is sampled via a $n_q = 3$ qubit digitization. This appears immediately relevant to the NS sampling theorem however as we will now discus, the HG functions do not quite fit the conditions for NS sampling and hence, some initial adjustments must be made before we can invoke the NS theorem.

As can be seen in FIG.5, the HG functions in field-space have infinite support, decaying exponentially to zero for large arguments. Furthermore the 'width'[14] of the functions increases with n. MSAH show that by truncating the support in field space, the first N HG functions can

[14] Defined by MSAH as "the interval range including most of the weight $|\psi_n(\phi)|$" [18]

be approximated up to an arbitrarily small error which decreases exponentially as the truncation value is increased. Specifically: for a cut-off value of $n,\ N$, an L can be defined s.t.

$$\forall\, n < N,\ \ |\psi_n(\phi)| < \epsilon \ \text{ for } \ |\phi| > L, \tag{38}$$

where $\epsilon \propto e^{-\frac{L^2}{2}}$ is exponentially small. (19) then implies that the exact same statement is true in conjugate-momentum space:

$$\forall\, n < N,\ \ \left|\widetilde{\psi}_n(k)\right| < \epsilon \ \text{ for } \ |k| > L, \tag{39}$$

where $\epsilon \propto e^{-\frac{L^2}{2}}$ gives the same error as before for the same L and N. In this way, we can define truncated versions of the first N HG functions which approximate the true HG functions up to an exponentially small error that scales with the truncation. These truncated functions now meet the requirements for NS sampling and hence by invoking the theorem, we can reconstruct them up to no loss of information by discrete sampling provided $\delta_\phi < \frac{\pi}{k_{max}}$, resulting in an overall error arising purely from the truncation of support. We now seek to minimise this error in pursuit of an optimal truncation value. We have that the error arising due to truncation in field space scales as $\epsilon_\phi \propto e^{-\frac{\bar{\phi}_{max}^2}{2}}$, decreasing exponentially as we increase our truncation value $\bar{\phi}_{max}$. However, as demonstrated by FIG.4, the fixed number of samples in our digitization means that by increasing $\bar{\phi}_{max}$, the sample spacing $\delta_{\tilde{\phi}}$ is also increased, shrinking the first Brillouin zone and hence the coverage of conjugate-momentum space achieved by our digitization in momentum space. This ultimately results in a reduction in the truncation of $\widetilde{\psi}_n(k)$ in conjugate-momentum space, corresponding to an increase in error due to under sampling in momentum space $\epsilon_k \propto e^{-\frac{k_{max}^2}{2}}$. We hence have a trade-off between sufficient coverage of field space and sufficient coverage of conjugate-momentum space. The obvious resolution is to ask the same level of accuracy in both representations, corresponding to equating the truncation value in both bases. We have

$$k_{max} = \frac{\pi}{\delta_{\tilde{\phi}}}\frac{n_s - 1}{n_s} = \frac{\pi(n_s-1)^2}{2\bar{\phi}_{max} n_s} \tag{40}$$

where the first equality follows from (23) with T.B.C.s and the second from the definition of $\delta_{\tilde{\phi}}$ (22). Now equating supports within our digitization framework, i.e. setting $k_{max} \equiv \bar{\phi}_{max}$ yields

$$\begin{aligned} \bar{\phi}_{max}^2 &= \frac{\pi}{2}\frac{(n_s-1)^2}{n_s} \\ \implies \bar{\phi}_{max} &= \sqrt{\frac{\pi}{2}\frac{(n_s-1)^2}{n_s}} \end{aligned} \tag{41}$$

in agreement with the formula proposed by BFN [23]. The convention chosen for the digitization of conjugate-momentum space should now make sense; the basis was set up such that under twisted boundary conditions our basis is symmetric about the zero mode in the same way that our $\tilde{\phi}$ basis is symmetric around $\bar{\phi} = 0$, allowing us to equate supports. Furthermore, under T.B.C.s the maximum momentum mode lies just within the first Brillouin zone, $k_{max} = \frac{n_s-1}{n_s} k_{\tilde{\phi}}^{max}$, such as to automatically saturate the NS bound:

$$\delta_{\tilde{\phi}} = \frac{\pi}{k_{\tilde{\phi}}^{max}} < \frac{\pi}{k_{max}} \tag{42}$$

We conclude from this discussion that the optimal field-space truncation value corresponds to saturation of the NS bound *in combination* with equivalence of field space and conjugate-momentum space coverage achieved by our digitization framework.

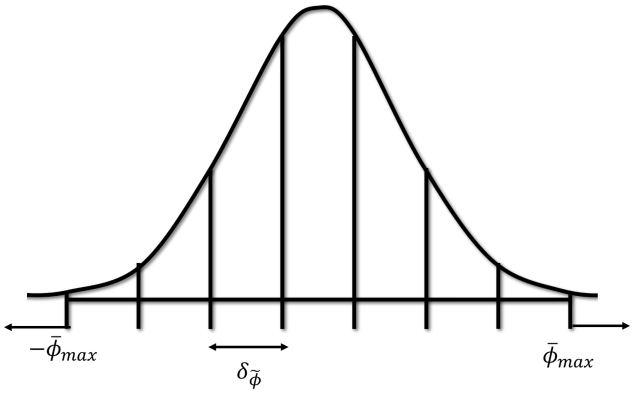


FIG. 4: Digitizing field space with $n_q = 3$ qubits amounts to discrete sampling of the $\tilde{\psi}_n(\phi)$ at $2^3 = 8$ evenly distributed points in field space. Since the number of samples $n_s = 2^{n_s}$ is fixed by the number of qubits, increasing the truncation value $\bar{\phi}_{max}$ increases the sample spacing $\delta_{\tilde{\phi}}$, shrinking the 1st Brillouin zone.

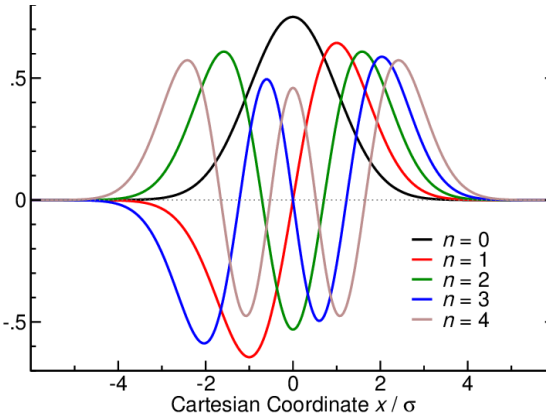


FIG. 5: The Hermite-Gauss (HG) functions of orders 0-4. The HG functions have infinite support however decay exponentially to zero for large arguments. Furthermore, the 'width'[14] of the functions increases with n. Source: [25].

We now check that this result aligns with the results of the investigation undertaken by KS. To do this we first reproduce Fig.2 of [22], which shows how percent error in calculated eigenenergies varies with field-space truncation value $\bar{\phi}_{max}$ for 3, 4 and 5 qubit digitizations of the HO. To do this, we define our digitized field-space and conjugate-momentum-space bases for general n_q then represent the digitized $\tilde{\phi}^2$ operator and the exact $\bar{\Pi}^2$ operator as diagonal $2^{n_q} \times 2^{n_q}$ square matrices acting in their respective bases. An inverse symmetric quantum Fourier transform is then performed on the the matrix representing $\bar{\Pi}^2$ in order to represent it in the digitized field-space basis. Now that the two operators are expressed as matrices in the same basis, they can be added together to compose the digitized HO Hamiltonian, $\tilde{H} = \frac{1}{2}\bar{\Pi}^2 + \frac{1}{2}\tilde{\phi}^2$, as a matrix acting in digitized field space from which the expected eigenenergies can be obtained via exact diagonalization. FIG.6 presents expected percent errors in ideal calculations of the first 5 eigenenergies, w.r.t field-space truncation $\bar{\phi}_{max}$ for $n_q = 3, 4$ and 5. The plot agrees exactly with Fig.2 of [22], up to a precision issue in the $n_q = 5$ case causing the results to break down at order 10^{-12} due to calculations being limited to double-precision. Fortunately, only the $n_q = 5$ plot appears to be affected, with the plots for $n_q = 3, 4$ in agreement with the theoretical discussion prior: the plot clearly shows the percent error initially decreasing as $\bar{\phi}_{max}$ increases due to improvements on the insufficient coverage in field space, before peaking at equality of supports and then increasing as the bound is surpassed, due to insufficient coverage in momentum space. The exponential improvement with n_q is also clearly visible (more qubits allow for a larger number of samples in our digitized regime, allowing greater overall coverage at equality of supports). The dashed lines correspond to optimal $\bar{\phi}_{max}$ values predicted by the formula (41) and we see that they are in clear agreement with the plot.

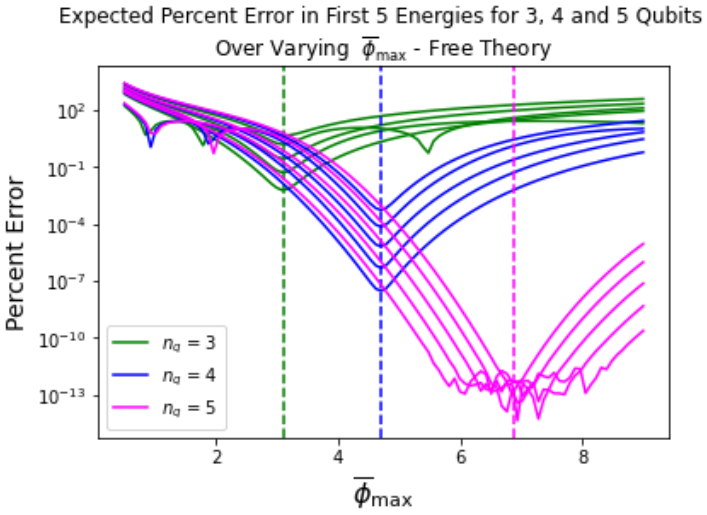


FIG. 6: The percent error expected for calculations of the lowest 5 HO energies via JLP digitization with field-space truncation $\bar{\phi}_{max}$ on ideal 3,4 and 5 qubit systems. The vertical, dashed lines correspond to predictions of (41) based on equality in the coverage of field space and of conjugate-momentum space.

3.3. The Harmonic Oscillator Basis

KS present an alternative approach to digitization which, unlike the JLP and MSAH algorithms, uses eigenstates of a harmonic oscillator with frequency ω_ϕ to digitize the field rather than eigenstates of $\bar{\phi}$. They assert that this is a viable approach since the NS sampling theorem places no restriction on the basis used to digitize the field, provided it spans the $\bar{\phi}$ space and the $\bar{\Pi}$ space of the lowest lying eigenstates to the desired level of accuracy. Such an approach appears potentially advantageous when considering systems with localized eigenfunctions which fall as a Gaussian at large distance and will be of particular interest when we come to consider Hamiltonian Truncation Effective Theory as an approach to digitization. We now introduce the HO basis construction and its corresponding qubit representation as presented by KS [22]:

The basis is defined as the first $n_s = 2^{n_q}$ eigenstates of a HO with frequency ω_ϕ. The Hamiltonian describing such a HO is

$$H_{basis} = \frac{1}{2}\bar{\Pi}^2 + \frac{1}{2}\omega_\phi^2\bar{\phi}^2 = \omega_\phi\left(a^\dagger_{\omega_\phi}a_{\omega_\phi} + \frac{1}{2}\right) \tag{43}$$

such that the digitized Hilbert space is defined:

$$\mathcal{H}_{dig} := span\{|n\rangle\}_{n<n_s} \quad \text{where} \quad H_{basis}\,|n\rangle = E_n\,|n\rangle\,. \tag{44}$$

The Hamiltonian describing the free theory (15) can then be expressed in terms of H_{basis} plus a correction term δH_{ω_ϕ}:

$$\bar{H} = \frac{1}{2}\bar{\Pi}^2 + \frac{1}{2}\bar{\phi}^2 = \underbrace{\frac{1}{2}\bar{\Pi}^2 + \frac{1}{2}\omega_\phi^2\bar{\phi}^2}_{H_{basis}} + \underbrace{\frac{1}{2}\left(1-\omega_\phi^2\right)\bar{\phi}^2}_{\delta H_{\omega_\phi}}\,. \tag{45}$$

Now upon digitizing the theory, equivalent to restricting to $\mathcal{H}_{dig}$, H_{basis} can, by definition, be represented as a diagonal matrix in the digitized basis, whilst δH_{ω_ϕ} can similarly be defined as a matrix in the digitized basis via raw truncation to $\mathcal{H}_{dig}$:

$$\left(\delta H_{\omega_\phi}\right)_{m,n} := \langle m|\,\delta H_{\omega_\phi}\,|n\rangle = \frac{1}{2}\left(1-\omega_\phi^2\right)\langle m|\,\bar{\phi}^2\,|n\rangle \tag{46}$$

where $m, n < n_s$. The raw truncation $\langle m| \bar{\phi}^2 |n\rangle$ is straightforwardly evaluated by expressing $\bar{\phi}^2$ in terms of creation/annihilation operators:

$$\bar{\phi}^2 = \frac{1}{2\omega_\phi} \left(a^\dagger_{\omega_\phi} + a_{\omega_\phi} \right)^2 \tag{47}$$

such that

$$\bar{\phi}^2 |n\rangle = \frac{1}{2\omega_\phi} \left[\sqrt{(n+2)(n+1)}\, |n+2\rangle + (2n+1)\, |n\rangle + \sqrt{n(n-1)}\, |n-2\rangle \right] \tag{48}$$

(since the $|n\rangle$ are eigenstates of H_{basis}) and using orthonormality of the basis states. A derivation of this formula as well as an example of its implementation can be found in Appendix E.

To evaluate the action of our operators on a system of qubits, we simply identify our basis states $|n\rangle$ with the computational basis states that span our qubit space, just as we did for the field-space and conjugate-momentum-space bases in the JLP formalism. The operators, now expressed as matrices acting in the computational basis, can then be decomposed into tensor products of single-qubit unitaries as described in Appendix D.[15]

In the JLP formalism, by tuning $\bar{\phi}_{max}$ we could directly tune the field-space coverage achieved by our digitized basis and hence, via $\delta_{\hat{\phi}}$, the coverage of conjugate-momentum space. In this way we were able to optimize the accuracy of our calculations. In the case of the HO basis we have that both the frequency ω_ϕ and number of basis states $n_s = 2^{n_q}$ jointly determine coverage of both field space and conjugate-momentum space. As such, by tuning ω_ϕ for fixed values of n_q we can optimise convergence of our basis states to the function being reconstructed and hence minimise the error in our calculations. For the free theory, the tuning of ω_ϕ is trivial: by setting $\omega_\phi = 1$, $\bar{H}$ is simply a diagonal matrix in the digitized basis and we can therefore (in theory) attain arbitrary accuracy in calculation of the lowest eigenenegies. In pursuit of a more interesting example we now progress to the case of a scalar field with quartic self-interaction.

4. DIGITIZATION- $\lambda\phi^4$ THEORY

Adding the quartic self-interaction term to our redefined HO Hamiltonian (15) yields

$$\bar{H} = \frac{1}{2}\bar{\Pi}^2 + \frac{1}{2}\bar{\phi}^2 + \frac{\bar{\lambda}_0}{4!}\bar{\phi}^4 \tag{49}$$

where for completeness, $\bar{\lambda}_0 = \frac{\hat{\lambda}_0}{\hat{m}_0}$. We now briefly outline how both the HO and JLP formulations are extended to accommodate this interaction term.

4.1. Extending The Harmonic Oscillator Basis

To extend the HO basis construction from the free theory to the interacting theory we simply add another operator, H_{int}, to our digitized Hamiltonian, defined as the raw truncation of the interaction term to the digitized Hilbert space $\mathcal{H}_{dig}$, just as we did for δH_{ω_ϕ}. We have:

$$\left(H_{int}\right)_{m,n} := \frac{\bar{\lambda}_0}{4!} \langle m| \bar{\phi}^4 |n\rangle \tag{50}$$

[15] Note that since δH_{ω_ϕ} is not diagonal, it has a more complex decomposition involving tensor products with mixed-Pauli gates.

which can be evaluated using the relation

$$\bar{\phi}^4 |n\rangle = \frac{1}{4\omega_\phi^2} \Big[\sqrt{(n+4)(n+3)(n+2)(n+1)}\, |n+4\rangle + (4n+6)\sqrt{(n+2)(n+1)}\, |n+2\rangle + (6n^2+6n+3)\, |n\rangle + (4n-2)\sqrt{n(n-1)}\, |n-2\rangle + \sqrt{n(n-1)(n-2)(n-3)}\, |n-4\rangle \Big] \tag{51}$$

defined in Appendix E (equivalent to applying (48) twice), in conjunction with orthonormality of the $|m\rangle, |n\rangle$. In full, we have

$$\bar{H}_{n,m} = \frac{\omega_\phi}{2}(2n+1)\delta_{n,m} + \frac{1}{2}\left(1-\omega_\phi^2\right) \langle n| \bar{\phi}^2 |m\rangle + \frac{\bar{\lambda}_0}{4!} \langle n| \bar{\phi}^4 |m\rangle \tag{52}$$

for $n, m < n_s$. Where (48),(51) are used to evaluate the overlaps (Appendix E).

4.2. Extending The JLP Basis

Extending the JLP operator structure is even more straightforward: we already have $\tilde{\phi}$ expressed as a diagonal matrix in the digitized field-space basis

$$\tilde{\phi} = diag\left(\tilde{\phi}^0, \tilde{\phi}^1, \ldots, \tilde{\phi}^{n_s-1}\right) \tag{53}$$

so the interaction term is

$$\frac{\bar{\lambda}_0}{4!}\tilde{\phi}^4 = \frac{\bar{\lambda}_0}{4!} diag\left(\left(\tilde{\phi}^0\right)^4, \left(\tilde{\phi}^1\right)^4, \cdots, \left(\tilde{\phi}^{n_s-1}\right)^4\right) \tag{54}$$

which we simply add on to our Hamiltonian. Note that this operator, like the $\tilde{\phi}^2$ operator is evaluated in *field* space.

4.3. Comparisons

In either case, we now have the digitized Hamiltonian for our self-interacting scalar field, expressed as a matrix in our digitized basis. As in the previous section, we can determine the low-lying eigenenergies expected from calculations on an ideal quantum computer by exact diagonalization of the matrix Hamiltonian. We can similarly plot the percent error in the calculated ground state energy over varied field-space truncation for fixed numbers of qubits to showcase how the parameters can be tuned for optimal accuracy in our calculations. In a similar effort to Fig.7 of KS's investigation [22], FIG.7 compares the results of such an effort for JLP digitization (dashed lines) with those of HO digitization (solid lines) for the strongly coupled ($\bar{\lambda}_0 = 32$) theory. In the case of JLP digitization, the field-space truncation variable is trivially $\tilde{\phi}_{max}$ as before, however in the HO regime, the equivalent quantity to $\tilde{\phi}_{max}$ is an expression of ω_ϕ and n_q, since both of these variables jointly determine the field-space coverage (see Appendix F for a derivation). FIG.7 agrees with Fig.7 of [22] up to the fact that our plots for HO digitization are smoother. It appears that KS attempted to take a shortcut in their investigation; after using (48) to derive the $\tilde{\phi}^2$ matrix, they defined $\tilde{\phi}^4$ as the square of the

resulting $\tilde{\phi}^2$ matrix. This differs from the approach taken here, where we first derived (51) by expressing $\tilde{\phi}^4$ in terms of creation/annihilation operators and used this to define the $\tilde{\phi}^4$ matrix. These two approaches yield different results since we are working in a truncated Hilbert space, and the derivation of (51) takes into account states above the cut-off which the first approach fails to account for. Nevertheless, the same conclusions can be drawn from both Fig.7 of [22] and FIG.7. We conclude that both bases provide digitization errors that for $n_q \geq 3$ are significantly smaller than those expected from computations on NISQ-era hardware. As such, the digitization of the scalar field is not expected to limit the accuracy of NISQ-era computations. There are some differences in the two approaches however: unlike the JLP approach which requires a symmetric QuFoTr of the qubit register, the action of the HO digitized Hamiltonian can be evaluated entirely in one basis. Furthermore, it appears the HO approach offers slightly improved accuracy for the correct choice of parameters when compared to the JLP algorithm. However, due to more complex operator structure, the HO approach requires significantly more resource requirements to implement, specifically in the number of gates and the complexity of gates, thus, it is perhaps less desirable when considering calculations on NISQ-era devices.

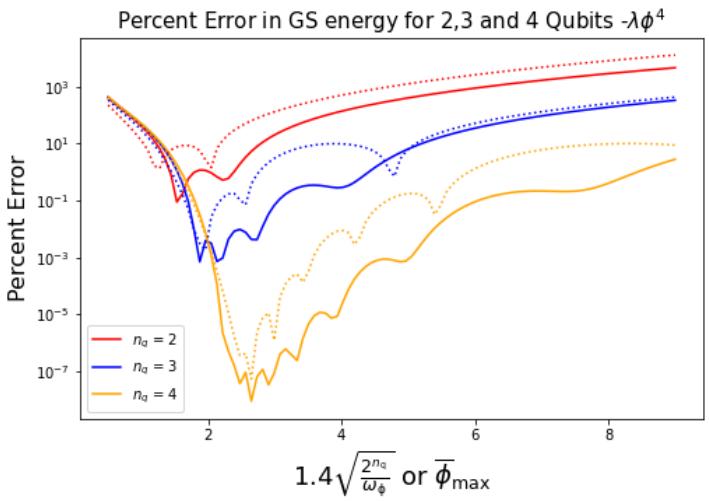


FIG. 7: Percent error in GS energy of (49) expected from calculations on ideal systems of 2,3 and 4 qubits via JLP digitization (dashed) and HO digitization (solid) w.r.t. field-space truncation. $\bar{\lambda}_0 = 32$ to represent strong coupling. The HO basis equivalent of $\bar{\phi}_{max}$ is $1.4\sqrt{\frac{2^{n_q}}{\omega_\phi}}$ (Appendix F). Expected GS energies are obtained via exact diagonalization of the digitized Hamiltonian expressed as a matrix in the digitized basis, and percent error is w.r.t the true GS energy.

5. HAMILTONIAN TRUNCATION EFFECTIVE THEORY

5.1. Overview

It is at this point that a sufficient understanding of digitization has been attained, at least for the case of quantum mechanics, and we finally turn our attention to Hamiltonian Truncation Effective Theory (HTET). In their 2022 paper [21], Cohen, Farnsworth, Houtz and Luty present HTET as a non-perturbative numerical method for calculating QFT observables based on truncation of the interacting Hamiltonian to a finite-dimensional effective Hilbert space. This should already seem relevant to our investigation into digitization where we were reconstructing Hamiltonians in finite-dimensional spaces. The following discussion offers a brief introduction to HTET as originally presented by CFHL, except we remain in the context of quantum mechanics. This will provide sufficient background context to support the main section of this

report, where HTET is applied to the HO digitization of $\lambda\phi^4$ theory in 0+1 dimensions.

The fundamental narrative of HTET involves taking our interacting Hamiltonian

$$H = H_0 + V, \tag{55}$$

and reconstructing it in an effective, finite-dimensional Hilbert space $\mathcal{H}_{eff}$, spanned by eigenstates of the free Hamiltonian H_0 up to some truncation energy E_T. Namely, we have

$$\mathcal{H}_{eff} := span\left\{ |i\rangle \right\}_{E_i \leq E_T} \quad \text{where} \quad H_0 |i\rangle = E_i |i\rangle . \tag{56}$$

When defining our reconstructed, effective Hamiltonian, the trivial approach is to simply take the raw truncation of the full Hamiltonian to $\mathcal{H}_{eff}$. Such that

$$\left(H_{eff}\right)_{i,j} := \langle i| H |j\rangle = \langle i| H_0 |j\rangle + \langle i| V |j\rangle \quad \text{for} \quad |i\rangle, |j\rangle \in \mathcal{H}_{eff} \tag{57}$$

HTET improves upon this trivial matching, defining the effective Hamiltonian order by order in powers of V

$$H_{eff} = H_0 + H_1 + H_2 + \cdots, \quad H_n = \mathcal{O}(V^n) \tag{58}$$

by matching a 'transition matrix' with the full theory. Here, the H_1 term corresponds to the trivial, raw truncation of V whilst the higher order corrections seek to factor back in information from the higher energy states that have been truncated away. We now define what is meant by the 'transition matrix' before presenting the subsequent matching conditions that define H_{eff}.

5.2. Defining The Transition Matrix

CFHL claim that the correct quantity to match for the effective Hamiltonian is a 'transition matrix'. This quantity is defined similarly to the S-matrix: we first turn off interaction adiabatically for $t > 0$ s.t.

$$V \to V e^{-\epsilon t}, \quad \epsilon > 0. \tag{59}$$

We then define the amplitude

$$\Sigma := \lim_{t_f \to \infty} U_{IP}\left(t_f, t_i = 0\right) \tag{60}$$

where U_{IP} is the interaction picture time evolution operator. Luty describes this quantity as comparable to "the square root of the S-matrix" since we only take t_f to infinity. It represents the amplitude to go from an interacting state at $t_i = 0$ to a free state as $t_f \to \infty$ and importantly, is well defined for normalizable states in finite volume.

A perturbative expansion can be constructed for Σ (Appendix G), the leading terms of which are given

$$\begin{aligned} \langle f|\Sigma|i\rangle = \delta_{fi} + \frac{\langle f|V|i\rangle}{E_{fi} + i\epsilon} + \sum_{\alpha} \frac{\langle f|V|\alpha\rangle \langle \alpha|V|i\rangle}{(E_{fi} + i\epsilon)(E_{f\alpha} + i\epsilon)} \\ + \sum_{\alpha,\beta} \frac{\langle f|V|\alpha\rangle \langle \alpha|V|\beta\rangle \langle \beta|V|i\rangle}{(E_{fi} + i\epsilon)(E_{f\alpha} + i\epsilon)(E_{f\beta} + i\epsilon)} + \mathcal{O}(V^4). \end{aligned} \tag{61}$$

where $E_{f\alpha} = E_f - E_\alpha$. For convenience the transition matrix T is then defined by

$$\langle f|\Sigma|i\rangle = \delta_{fi} + \frac{\langle f|T|i\rangle}{E_{fi} + i\epsilon}, \tag{62}$$

to factor out the E_{fi} denominator. We hence get that T has the perturbative expansion

$$\langle f|T|i\rangle = \langle f|V|i\rangle + \sum_\alpha \frac{\langle f|V|\alpha\rangle \langle \alpha|V|i\rangle}{(E_{f\alpha} + i\epsilon)} + \sum_{\alpha,\beta} \frac{\langle f|V|\alpha\rangle \langle \alpha|V|\beta\rangle \langle \beta|V|i\rangle}{(E_{f\alpha} + i\epsilon)(E_{f\beta} + i\epsilon)} + \mathcal{O}(V^4). \tag{63}$$

This is the quantity that is matched between the finite-dimensional effective theory and the full theory to derive the matching conditions we use to define our effective Hamiltonian H_{eff} (see Appendix H for full derivation).

5.3. The Matching Conditions

As described in Appendix H, matching the transition matrix in the effective theory, T_{eff}, to the transition matrix in the full theory, T, to 3rd order yields the matching conditions

$$\begin{aligned}
\langle f|H_1|i\rangle_{eff} &= \langle f|V|i\rangle \\
\langle f|H_2|i\rangle_{eff} &= \sum_\alpha^{>} \frac{\langle f|V|\alpha\rangle \langle \alpha|V|i\rangle}{E_{f\alpha}} \\
\langle f|H_3|i\rangle_{eff} &= \sum_{\alpha,\beta}^{>} \frac{\langle f|V|\alpha\rangle \langle \alpha|V|\beta\rangle \langle \beta|V|i\rangle}{E_{f\alpha}E_{f\beta}} - \sum_\alpha^{<} \sum_\beta^{>} \frac{\langle f|V|\alpha\rangle \langle \alpha|V|\beta\rangle \langle \beta|V|i\rangle}{E_{\alpha\beta}E_{f\beta}}
\end{aligned} \tag{64}$$

which we use to define our effective Hamiltonian $H_{eff} = H_0 + H_1 + H_2 + H_3 + \mathcal{O}\left(V^4\right)$.

6. THE HARMONIC OSCILLATOR BASIS WITH 2ND ORDER IMPROVEMENT

6.1. The Harmonic Oscillator Basis as 1st Order HTET

The 1st order matching condition in (64) corresponds to the raw truncation of the interaction term V to the finite-dimensional basis of H_0 eigenstates which span $\mathcal{H}_{eff}$. Thus, it is not difficult to see that the HO basis digitized Hamiltonian described by Klco and Savage [22] and discussed earlier in this paper, is analogous to the effective Hamiltonian, H_{eff}, described by HTET to 1st order. To see this we examine the HO basis approach to digitizing the interacting theory. The Hamiltonian is broken up into three terms:

$$\bar{H} = \frac{1}{2}\bar{\Pi}^2 + \frac{1}{2}\bar{\phi}^2 + \frac{\bar{\lambda}_0}{4!}\bar{\phi}^4 = \underbrace{\frac{1}{2}\bar{\Pi}^2 + \frac{1}{2}\omega_\phi^2\bar{\phi}^2}_{H_{basis}} + \underbrace{\frac{1}{2}\left(1 - \omega_\phi^2\right)\bar{\phi}^2}_{\delta H_{\omega_\phi}} + \underbrace{\frac{\bar{\lambda}_0}{4!}\bar{\phi}^4}_{H_{int}}. \tag{65}$$

The basis states spanning the digitized Hilbert space $\mathcal{H}_{dig}$ are then taken as the first $n_s = 2^{n_q}$ eigenstates of H_{basis}. By relabelling $H_{basis} \to H_0$ we see that the effective Hilbert space of HTET, $\mathcal{H}_{eff} := \{|n\rangle\}_{E_n < E_T} \equiv \mathcal{H}_{dig}$ when $E_T \equiv E_{n_s}$.

In the HO basis formalism, the δH_{ω_ϕ} and H_{int} terms are then represented as matrices acting in the digitized basis via raw truncation, i.e.

$$\left(\delta H_{\omega_\phi}\right)_{m,n} + \left(H_{int}\right)_{m,n} = \langle m| \, \delta H_{\omega_\phi} \, |n\rangle + \langle m| \, H_{int} \, |n\rangle \,, \quad n, m < n_s. \tag{66}$$

Looking at the 1st matching condition in (64) we see that these two terms collectively correspond to the first order term in our HTET efective hamiltonian, H_1, when $\delta H_{\omega_\phi} + H_{int}$ is treated as a collective intraction term:

$$\begin{aligned} \langle m| \, H_1 \, |n\rangle_{eff} := \langle m| \, V \, |n\rangle &= \langle m| \, \delta H_{\omega_\phi} + H_{int} \, |n\rangle \,, \quad m, n \in \mathcal{H}_{eff} \\ &\equiv \langle m| \, H_{\omega_\phi} \, |n\rangle + \langle m| \, H_{int} \, |n\rangle \,, \quad m, n < n_s \\ &= \left(\delta H_{\omega_\phi}\right)_{m,n} + \left(H_{int}\right)_{m,n} . \end{aligned} \tag{67}$$

Adopting the HTET notation, we hence have

$$\bar{H} = \frac{1}{2}\bar{\Pi}^2 + \frac{1}{2}\bar{\phi}^2 + \frac{\bar{\lambda}_0}{4!}\bar{\phi}^4 = \underbrace{\frac{1}{2}\bar{\Pi}^2 + \frac{1}{2}\omega_\phi^2\bar{\phi}^2}_{H_0} + \underbrace{\frac{1}{2}\left(1 - \omega_\phi^2\right)\bar{\phi}^2 + \frac{\bar{\lambda}_0}{4!}\bar{\phi}^4}_{V} \tag{68}$$

with our digitized/effective Hilbert space defined

$$\mathcal{H}_{dig} := \{|n\rangle\}_{E_n < E_{n_s}} \,, \quad \text{where} \quad H_0 \, |n\rangle = E_n \, |n\rangle \tag{69}$$

The digitized Hamiltonian acting in $\mathcal{H}_{dig}$ defined by the matching conditions is then given $H_{dig} \equiv H_{eff} = H_0 + H_1 + H_2 + H_3 + \mathcal{O}\left(V^4\right)$ which to first order, is defined by the matrix:

$$\begin{aligned} \left(H_{dig}\right)_{m,n} &= E_n\delta_{m,n} + \langle m| \, V \, |n\rangle \\ &= E_n\delta_{m,n} + \frac{1}{2}\left(1 - \omega_\phi^2\right)\langle m| \, \bar{\phi}^2 \, |n\rangle + \frac{\bar{\lambda}_0}{4!}\langle m| \, \bar{\phi}^4 \, |n\rangle \end{aligned} \tag{70}$$

which can be evaluated using (48) and (51) (Appendix E). We have hence shown that the HO basis digitized Hamiltonian described by KS corresponds to the effective Hamiltonian described by HTET evaluated to 1st order. It at this point seems reasonable to propose that adding higher order terms to our digitized (effective) Hamiltonian should offer improved convergence of our digitized theory to the true theory, allowing us to leverage higher accuracy in our calculations for a given number of qubits. The remainder of this paper investigates this proposition for the case of $\lambda\phi^4$ theory in 0+1 dimensions. We begin by deriving the corresponding 2nd order term for our digitized Hamiltonian as an expression of matrices acting in the digitized HO basis, such that expected accuracy in calculations can be determined by exact diagonalization, and resource requirements evaluated through decomposition into multi-Pauli gates.

6.2. Deriving 2nd Order Terms for the HO Basis Digitized Hamiltonian

From (64), the second order term in the effective Hamiltonian, H_2, is defined by the matching condition

$$\langle m|H_2|n\rangle_{eff} = \sum_{\alpha}^{>} \frac{\langle m|V|\alpha\rangle \, \langle \alpha|V|n\rangle}{E_{m\alpha}}, \tag{71}$$

where the summation is over eigenstates *above* the truncation energy. Expanding out the interaction term, $V = \frac{1}{2}\left(1-\omega_\phi^2\right)\bar{\phi}^2 + \frac{\bar{\lambda}_0}{4!}\bar{\phi}^4$, yields four terms:

$$\begin{aligned}
&\langle m|\, H_2\, |n\rangle_{eff} = \\
&\sum_\alpha^{>} \frac{\left(\frac{1}{2}\left(1-\omega_\phi^2\right)\langle m|\,\bar{\phi}^2\,|\alpha\rangle + \frac{\bar{\lambda}_0}{4!}\langle m|\,\bar{\phi}^4\,|\alpha\rangle\right)\left(\frac{1}{2}\left(1-\omega_\phi^2\right)\langle \alpha|\,\bar{\phi}^2\,|n\rangle + \frac{\bar{\lambda}_0}{4!}\langle \alpha|\,\bar{\phi}^4\,|n\rangle\right)}{E_{m\alpha}} \\
&= \underbrace{\frac{1}{4}\left(1-\omega_\phi^2\right)^2 \sum_\alpha^{>} \frac{\langle m|\,\bar{\phi}^2\,|\alpha\rangle\langle \alpha|\,\bar{\phi}^2\,|n\rangle}{E_{m\alpha}}}_{\langle m|\, H^{(2)}_{\bar{\phi}^2\bar{\phi}^2}\,|n\rangle_{eff}} + \underbrace{\left(\frac{\bar{\lambda}_0}{4!}\right)^2 \sum_\alpha^{>} \frac{\langle m|\,\bar{\phi}^4\,|\alpha\rangle\langle \alpha|\,\bar{\phi}^4\,|n\rangle}{E_{m\alpha}}}_{\langle m|\, H^{(2)}_{\bar{\phi}^4\bar{\phi}^4}\,|n\rangle_{eff}} \\
&\quad + \underbrace{\frac{1}{2}\left(1-\omega_\phi^2\right)\cdot\frac{\bar{\lambda}_0}{4!}\sum_\alpha^{>} \frac{\langle m|\,\bar{\phi}^2\,|\alpha\rangle\langle \alpha|\,\bar{\phi}^4\,|n\rangle}{E_{m\alpha}}}_{\langle m|\, H^{(2)}_{\bar{\phi}^2\bar{\phi}^4}\,|n\rangle_{eff}} + \underbrace{\frac{1}{2}\left(1-\omega_\phi^2\right)\cdot\frac{\bar{\lambda}_0}{4!}\sum_\alpha^{>} \frac{\langle m|\,\bar{\phi}^4\,|\alpha\rangle\langle \alpha|\,\bar{\phi}^2\,|n\rangle}{E_{m\alpha}}}_{\langle m|\, H^{(2)}_{\bar{\phi}^4\bar{\phi}^2}\,|n\rangle_{eff}}
\end{aligned} \tag{72}$$

Whilst all four of these terms involve sums over states *above* the cut-off, i.e. an infinite number of states, we find that when using (48) and (51) to evaluate the action of $\bar{\phi}^2$ and $\bar{\phi}^4$, only the first 2 to 4 states above the cut-off provide non-zero overlaps. By applying (48) and (51) and evaluating these non-zero overlaps, each term can be expressed as a matrix acting in the digitized basis analogously to the 1st order case discussed in Appendix E. Appendix I provides a detailed demonstration of this process, and presents the resulting matrices for the $n_q = 3$ case. The resulting four matrices collectively compose the second order correction to our digitized Hamiltonian and can be added to our existing digitized Hamiltonian matrix to give the HO basis digitized (effective) Hamiltonian to second order. Eigenenergies expected from calculations on an ideal quantum computer via this digitization framework can be obtained from the matrix Hamiltonian by exact diagonalization for general n_q, so that accuracy improvements compared to the first order case can be assessed. Before discussing the results however, we briefly digress to evaluate the additional resources required to implement such a 2nd order term.

6.3. Resource Requirements

Now that we have the second order term expressed as a collection of matrices acting in the digitized basis, the circuit construction required to evaluate its time-evolution can be determined by via a decomposition into multi-Pauli gates. This process is unchanged from the discussions in section 3.2.iii and Appendix D, where the decomposition is achieved one coefficient at a time. Undertaking this process by hand is long and tedious however, and quite unreasonable for $n_q > 2$. Instead, the decomposition can be achieved in Python where the number of *k*-body operators required for an n_q qubit digitization can be quickly evaluated. In this way the additional gate requirements of the 2nd order terms can be easily determined. The operator counts for one first-order-Trotterized step of time evolution [24] for both 1st and 2nd order HO basis digitized, $\lambda\phi^4$ Hamiltonians are presented in TABLE I for 2,3,4 and 5 qubits. In analogy with Table *ii* of

KS's investigation [22], total expected CNOT counts are also evaluated.[16] CNOT gates are of particular interest as they are a significant source of noise on NISQ-era hardware.

Digitization	n_q	0-body	1-body	2-body	3-body	4-body	5-body	$\text{CNOT}_{\text{uncompiled}}$
	2	1	3	2				4
HO basis	3	1	5	9	4			34
1st order	4	1	6	16	18	10		164
	5	1	7	22	32	44	22	612
	2	1	4	3				6
HO basis	3	1	6	11	6			46
2nd order	4	1	7	19	25	12		210
	5	1	8	26	48	53	24	754

TABLE I: Gate counts required for the circuit implementing one first-order-Trotterized step of time evolution [24] of the Hamiltonian describing $\lambda\phi^4$ scalar field theory in 0+1 dimensions, digitized via the HO basis at both 1st and 2nd order. Counts are given for 2, 3, 4 and 5 qubit digitizations, and the term k-body operator relates to a tensor product of k Pauli gates acting on the qubit space. Uncompiled CNOT counts are based on the upper bound of $2(k-1)$ CNOT gates for each $k > 1$ k-body operator.

Upon inspection of TABLE I, we see that the segment of the table corresponding to the 1st order digitization agrees exactly with Table *ii* of [22], confirming that our Python code is correct. The main observation however is that the additional resource requirements for the second order terms are surprisingly modest ($\sim 28\%$ additional CNOT gates for $n_q = 4$) considering the complicated operator structure involved. This is because the order of implementation of the multi-Pauli gates in the decomposition does not matter and hence, any multi-Pauli gates in the decomposition of H_2 that also appear in the decomposition of H_0+H_1 simply amount to a phase re-tuning of existing gates. In summary, only *new* combinations of Pauli operators contribute to the additional resource requirements. As discussed by Klco and Savage, adjacent pairs of CNOT operations can be cancelled and hence, the uncompiled counts given here serve as upper bounds to the actual, recompiled CNOT counts. In conclusion, the additional resource requirements of the 2nd order improvement term look promising when comparing to the 1st order digitization; such modest increases in gate requirements means that *any* potential gains in accuracy yielded by the implementation of our 2nd order terms will be of significance. Nevertheless, gate counts and CNOT counts for both 1st and 2nd order HO basis digitization are still significantly higher than those for the JLP digitization and we conclude that unless the second order improvement yields significantly improved accuracy compared with the first order case, JLP digitization still remains the preferable choice for NISQ-era devices, despite its inferior accuracy.

6.4. Results

Motivated by the earlier comparison of the JLP vs HO (1st order) bases, we begin by comparing expected percent error in GS energy calculations on ideal systems of n_q qubits, comparing the results for both the raw truncated (1st order) and 2nd order HO digitized Hamiltonians for a

[16] Each k-body operator for $k > 1$ requires $2(k-1)$ CNOT gates to implement.

scalar field with quartic self-interaction in 0+1 dimensions (49). Since we are not comparing with the JLP basis here, we plot w.r.t. frequency ω_ϕ rather than field-space truncation for convenience. In their work [21], CFHL claim that HTET works for both weak and strong coupling, with the exact diagonalisation of the effective Hamiltonian yielding errors in GS energy of order $1/E_T^3$ for the 2nd order case compared with $1/E_T^2$ for the 1st order case. FIG.8 shows the results for $\bar{\lambda}_0 = 32$ in line with the previous plots to represent strong coupling, for 2,3 and 4 qubit digitizations. Immediately we see that the addition of the second order 'improvement' actually appears to give *unfavourable* results: the 2nd order plots (dashed) are significantly less 'smooth' than the 1st order (solid) plots, with sharp fluctuations and, with the exception of the $n_q = 2$ case, actually yield *decreased* accuracy for most values of ω_ϕ. The sharp upwards spikes are due to select values of ω_ϕ in our matrix coefficients resulting in large changes to the digitized Hamiltonian matrix, whilst downwards spikes are due to the expected GS energy plot crossing the actual GS energy value as in the JLP case.[17] Again, these spikes dont occur for the 1st order case which sits consistently above the actual energy. One may propose that this behaviour indicates improved convergence to the actual value for certain ω_ϕ values, however taking advantage of such spikes would require extremely fine tuning of gate phases rendering them meaningless in practice and we will hence choose to ignore them. Furthermore, we see no evidence for the $\mathcal{O}\left(1/E_T^3\right)$ error scaling described by CFHL. Considering that $E_T \equiv \omega_\phi \left(2^{n_q} - 1/2\right)$, this should be visible as n_q increases. It would be interesting to investigate higher numbers of qubits here, as the E_T values attained here are in the lower range of those investigated by CFHL [21]. Note however that considering the complex circuit construction required for HO basis digitization, only systems with $n_q < 5$ qubits are realistically attainable in the NISQ-era.

Still in pursuit of validating evidence, we turn to the case of weak coupling. FIG.9 presents an investigation analogous to that of FIG.8 except with $\bar{\lambda}_0 = 2.4$ to represent weak coupling. Only the $n_q = 2$ and $n_q = 3$ cases are plotted due to insufficient precision in the actual GS energy value used to calculate percent errors. Given more time, the actual value could be determined to more significant figures such that plots for $n_q > 3$ could be included. Nevertheless FIG.9 tells a similar story to FIG.8 and we ultimately draw the same conclusions. It appears the coupling strength is not the issue here.

We now question whether our implementation of the 2nd order effective Hamiltonian is theoretically sound; in hindsight, by adopting the HO basis approach, the H_{basis} defining our HO basis states, which is taken as H_0 when comparing to HTET, doesn't actually correspond to the free scalar field theory associated with our interacting theory (15). This is the trick of the HO basis construction: an ω_ϕ dependence is introduced into our HO basis states via the introduction of a correction term δH_{ω_ϕ} such that the ω_ϕ dependence cancels in the full theory (65). We can alternatively define our effective Hamiltonian such that H_0 corresponds to (15), the free scalar field theory where $\omega_\phi = 1$, and there is now no need for the δH_{ω_ϕ} term that was previously treated as an interaction term. This is achieved by simply setting $\omega_\phi = 1$ in our current formalism, sending all of the unrelated coefficients to zero. The result is the second order effective Hamiltonian applied exactly to our full theory Hamiltonian (49), with H_0 now in agreement with the appropriate free scalar field Hamiltonian. In doing this we lose the ability to tune the frequency of our basis states, so the resulting percent errors are presented in tabular format. TA-

[17] Plotting the expected energy yields a plot that oscillates between sitting above and below the actual GS energy. If FIG.8 was produced with sufficient resolution, the dips would continue down towards zero.

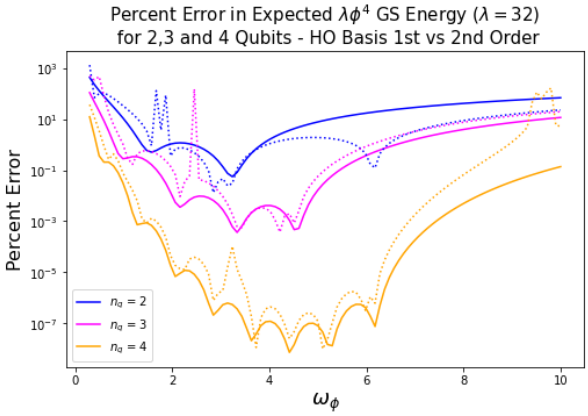


FIG. 8: Percent error expected in ideal calculation of GS energy via 1st order (solid) and 2nd order (dashed) HO basis digitizations with $n_q = 2, 3, 4$ qubits, w.r.t. basis frequency ω_ϕ. The target theory is $\lambda\phi^4$ scalar field theory with $\bar{\lambda}_0 = 32$ to represent strong coupling. Expected GS energy is obtained through exact diagonalization of the digitized Hamiltonian matrix.

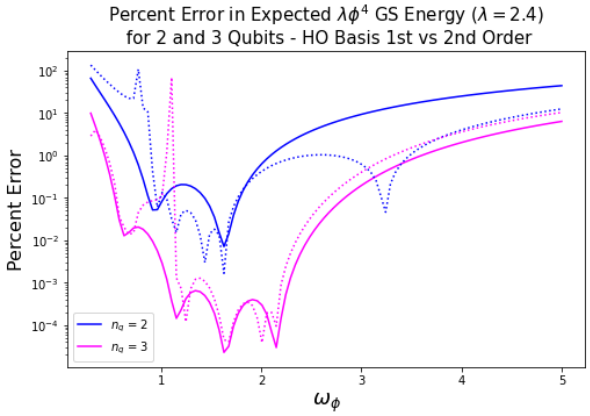


FIG. 9: The same investigation portrayed in FIG.8 applied here to the weakly coupled theory with $\bar{\lambda}_0 = 2.4$. We were unable to investigate the case of $n_q = 4$ due to only knowing the actual GS energy value to 6 decimal places. ω_ϕ denotes the frequency of the HO used to define the digitization basis states.

BLE II presents the percent errors for both strongly and weakly coupled $\lambda\phi^4$ theories digitized at both 1st and 2nd order with $\omega_\phi = 1$. Again, we see immediately that something is wrong: the percent errors are larger for the second order digitization, with the $\mathcal{O}(1/E_T^3)$ scaling described by CFHL [21] nowhere to be seen. Again, more data would be useful here, as we are still only just entering the lower range of E_T values explored in [21], but at least for the practical application of HTET to our specific QC context, where n_q must remain low, the addition of the second order 'improvement' appears to have no benefit thus far.

Digitization	$\bar{\lambda}_0$	$n_q = 2$	$n_q = 3$	$n_q = 4$
HO basis	2.4 (W)	0.075	0.0035	-
1st order	32 (S)	11	0.28	0.028
HO basis	2.4 (W)	0.13	0.1	-
2nd order	32 (S)	33	0.93	0.069

TABLE II: Percent errors expected in ideal calculations of GS energy via 1st order and 2nd order HO basis digitizations with $\omega_\phi = 1$ and $n_q = 2, 3, 4$ qubits, for $\lambda\phi^4$ scalar field theories with $\bar{\lambda}_0 = 32, 2.4$ to represent both strong and weak coupling. Again, the $n_q = 4$ case is omitted for weak coupling due to insufficient precision in the knowledge of the true energy value.

In search of more insight, we examine the simpler case of the free scalar field theory digitized via the HO basis. This is the set-up investigated by Klco and Savage [22] and discussed here in Section 3.3. Within our existing construction, this simply corresponds to setting $\bar{\lambda}_0 = 0$, resulting in just one 2nd order term, $H_2 = H^{(2)}_{\phi^2\phi^2}$. As discussed previously, by tuning $\omega_\phi = 1$, our basis states become the eigenstates of the target Hamiltonian, $\delta H\omega_\phi$ goes to zero and our digitized Hamiltonian is simply a diagonal matrix, yielding perfect accuracy (in theory) in expected, ideal eigenenergy calculations. For the free theory we have exact analytical solutions for the actual eigenenergies, so we can extend our investigation to higher energy states. FIG.10

shows the expected percent error for E_0, E_1, E_5, E_6 with a 3 qubit digitization for values of $\omega_\phi > 1.2$. These energy states have been chosen such that we have a representation of the lowest eigenstates and the higher accessible eigenstates (a 3-qubit digitization only gives us access to the first $2^3 = 8$ eigenenergies) without cluttering the plot. As expected, we see that for all eigenenergies, expected percent errors tend to zero as ω_ϕ approaches 1, with the 1st order digitization outperforming the 2nd order digitization for E_0 and E_1 in line with previous results. However, for E_5 and E_6 we see that the 2nd order digitization appears to offer a significant advantage over the raw case; the first indication of the 2nd order term providing a favourable outcome. Whilst there is little practical significance to this advantage, there is theoretical insight to be extracted. The fact that only the higher accessible energies were improved makes sense: the eigenstates of our target system are similar to those composing our basis, differing only by the value of ω_ϕ, and H_2 involves only states close to the truncation energy. It makes sense that improvements arising through such a correction would favour the higher energy target states. This idea also aligns with FIG.8 and FIG.9 for the interacting theory where the 2nd order digitization appears more effective for $n_q = 2$ qubits: in such a set-up, the target GS is closer to the highest accessible eigenstate such that the H_2 matrix affects the upper left corner of the overall digitized Hamiltonian, not just the lower right hand corner. Nevertheless, CFHL showed that HTET works for the lowest lying eigenstates [21], using the GS excitation energy $\Delta E_1 = E_1 - E_0$ to demonstrate the $\mathcal{O}\left(1/E_T^3\right)$ scaling. Motivated by FIG.10 it would perhaps be interesting to investigate how HTET performs for higher energy eigenstates and how this translates into the investigation undertaken here.

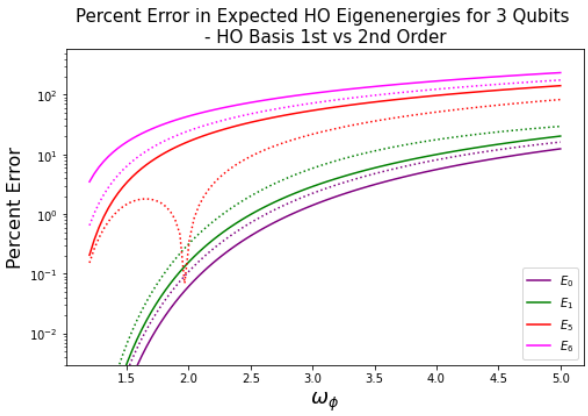


FIG. 10: Expected percent error for ideal calculation of E_0, E_1, E_5, E_6 in the free scalar field theory with a 3 qubit HO basis digitization at both 1st and 2nd order for values of $\omega_\phi > 1.2$.

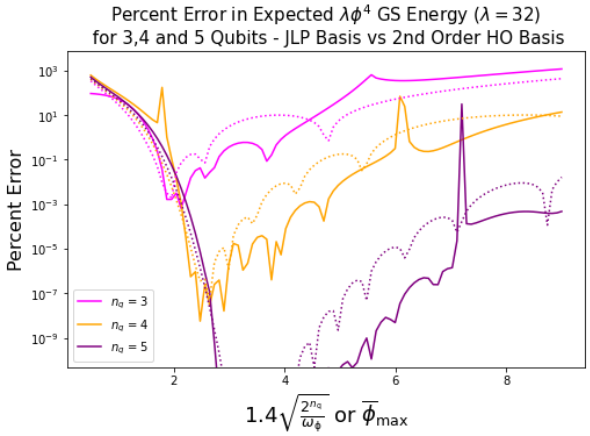


FIG. 11: Expected percent errors in GS energy for $\lambda\phi^4$ theory with 2nd order HO (solid) and JLP (dashed) digitizations. HO plots are w.r.t. the $\bar{\phi}_{max}$ equivalent variable and the error axis is truncated at 10^{-10} due to limitations of double-precision.

Due to time constraints, we conclude this investigation with one final plot; FIG.11 compares expected percent error in strongly-coupled $\lambda\phi^4$ GS energy for calculations via JLP digitization vs 2nd order HO digitization. The HO basis plots are with respect to the equivalent field-space truncation variable $1.4\sqrt{2^{n_q}/\omega_\phi}$ discussed in Appendix F. The purpose of this plot is to demonstrate that whilst the 2nd order HO digitization constructed here appears inferior to the raw

digitized case, it still outperforms JLP digitization. We ultimately draw the same conclusions as Klco and Savage [22] however, that despite it's inferior accuracy in calculations, the JLP basis compensates with significantly less noise potential due to its efficient use of resources compared to any HO basis and HTET approach to digitization. It is important to remember that all plots presented in our investigation concern expected calculations on *ideal* quantum devices where noise is not considered, and hence in the case of a real, practical application on NISQ-era devices, the JLP basis is certainly more favourable for the digitization of scalar field theories.

7. CONCLUSIONS

Summarising the narrative of our investigation, we began with a preliminary, supporting investigation into the established literature on digitization of scalar field theories, building on the work of Klco and Savage [22] that unified the established JLP algorithm with the insights explored by MSAH [18][19] relating to the NS sampling theorem. We worked alongside this investigation, filling in gaps and contributing insights using related literature; a formula proposed by BFN [23] for optimal field-space truncation in the JLP digitized free scalar field theory was derived in 0+1 dimensions using insights from MSAH [18], and shown to agree with the findings of KS [22]. We then explored the HO digitization basis proposed by KS and compared it to the JLP basis approach for strongly coupled $\lambda\phi^4$ scalar field theory in 0+1 dimensions. Upon building a sufficient foundational understanding of the digitization of scalar field theories in 0+1 dimensions, we examined the Hamiltonian truncation effective theory proposed by CFHL [21], remaining in the context of QM, and identified the HO basis digitization described by KS [22] as analogous to 1st order HTET. Motivated by this observation, a 2nd order 'improvement' term was added to the HO basis digitized Hamiltonian, aiming to improve the scaling in digitization error of ideal eigenenergy calculations by $\mathcal{O}\left(1/E_T\right)$. Whilst evaluation of additional resource requirements found that our second order term only required $\sim 28\%$ more CNOT gates compared with the raw digitized theory for $n_q = 4$ qubits, exact diagonalization of resulting digitized Hamiltonians, for both the free and quartic, self-interacting scalar field theories, found that our proposed implementation of a 2nd order improvement term was ultimately unsuccessul, proving mostly inferior to the raw digitized case. Although still providing superior performance over the JLP basis for the given systems digitized on *ideal* devices, we conclude that due to resource requirements and the resulting noise landscape, the JLP basis remains the favourable choice for digitization of scalar field theories in the NISQ-era.

8. FUTURE WORK

This investigation was cut short due to time constraints; given more time, further exploration would be undertaken on several fronts. Firstly, a deeper investigation into the weakly coupled system would be of interest, with more precise true eigenenergy values. Also, an investigation into the implications of our 2nd order term (as well as HTET in general) on the error in reconstructed eigenenergies of higher energy, i.e. closer to E_{n_s}, for $\lambda\phi^4$ theory could prove potentially insightful. Finally, exploration of higher dimensional theories through the addition of more lattice sites represents an optimistic vision for the future.

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SUMMARY FOR A GENERAL AUDIENCE

Quantum Field Theory (QFT) is the framework used by physicists to describe the behaviour of particles and fields at the smallest scales of the universe. Whilst QFT is generally a successful theory, it often involves calculations that are extremely long and complicated. To get around this, scientists turn to computers to speed up the process. QFT can be so complicated however that even classical computers limit our exploration.

Quantum computers offer a new, cutting-edge approach to computing; they work with extremely small particles called 'qubits', allowing the computer to utilize strange quantum phenomena such as 'superposition' whereby qubits can exist in two states at once, and 'entanglement' where qubits can become interconnected. These phenomena allow quantum computers to explore more complicated problems than classical computers, and at faster speeds.

Quantum computers are still in early stages of development but may be more suited to performing QFT calculations than classical computers. The research in this paper concerns how a QFT may be reconstructed on such a digital quantum computer, by sampling the theory at specific points in space (discretization) and constraining the field to take a finite set of specific values (digitization). In this way, the infinite dimensional QFT can be approximated in finite dimensions. This paper explores various methods to achieve the 'digitization' process.

STATEMENT OF GENERATIVE AI USE

ChatGPT 3.5 was used for assistance with writing both Python and LaTeX code. It was mostly asked to explain how certain commands and functions are used/implemented. In this sense, it was treated as a useful learning resource rather than a shortcut.

APPENDICES

A. Finite-Difference Approximation for the Discretized Laplacian Operator

In d dimensions, the Laplacian operator acts on the field $\phi(\boldsymbol{x})$ as

$$\boldsymbol{\nabla}^2\phi(\boldsymbol{x}) = \sum_{i=1}^{d} \frac{\partial^2}{\partial x_i^2}\phi(\boldsymbol{x}) \tag{73}$$

and partial derivatives are defined

$$\frac{\partial}{\partial x_i}\phi(\boldsymbol{x}) = \lim_{\delta x \to 0} \frac{\phi\left(\boldsymbol{x} + \hat{\boldsymbol{\mu}}_i \delta x\right) - \phi(\boldsymbol{x})}{\delta x}. \tag{74}$$

Now, in our discretized regime where δx is fixed and $\phi(\boldsymbol{x}) \to \phi_{\boldsymbol{x}}$, the best finite-difference approximation turns out to be the **central difference approximation**:

$$\frac{\partial}{\partial x_i}\phi_{\boldsymbol{x}} \approx \frac{\delta_{\delta x}\left[\phi\right]_{\boldsymbol{x}}}{\delta x} \tag{75}$$

where

$$\delta_{\delta x}\left[\phi\right]_{\boldsymbol{x}} := \phi_{\boldsymbol{x}+\frac{\delta x}{2}\hat{\boldsymbol{\mu}}_i} - \phi_{\boldsymbol{x}-\frac{\delta x}{2}\hat{\boldsymbol{\mu}}_i} \tag{76}$$

defines the **central difference**.

Notice that here we have have half-spacings which are not feasible in our lattice definition, however we will now iterate this approximation once to obtain an approximation for $\frac{\partial^2}{\partial x_i^2}\phi_{\boldsymbol{x}}$ involving spacings of only $\pm\delta x$. We have

$$\frac{\partial^2}{\partial x_i^2}\phi_{\boldsymbol{x}} \approx \frac{\delta_{\delta x}\left[\frac{\partial}{\partial x_i}\phi\right]_{\boldsymbol{x}}}{\delta x}. \tag{77}$$

(76) implies

$$= \frac{1}{\delta x}\left(\left[\frac{\partial}{\partial x_i}\phi\right]_{\boldsymbol{x}+\frac{\delta x}{2}\hat{\boldsymbol{\mu}}_i} - \left[\frac{\partial}{\partial x_i}\phi\right]_{\boldsymbol{x}-\frac{\delta x}{2}\hat{\boldsymbol{\mu}}_i}\right), \tag{78}$$

then (75) followed by (76) yields

$$\begin{aligned} &\approx \frac{1}{\delta x}\left(\frac{\delta_{\delta x}\left[\phi\right]_{\boldsymbol{x}+\frac{\delta x}{2}\hat{\boldsymbol{\mu}}_i}}{\delta x} - \frac{\delta_{\delta x}\left[\phi\right]_{\boldsymbol{x}-\frac{\delta x}{2}\hat{\boldsymbol{\mu}}_i}}{\delta x}\right) \\ &= \frac{1}{(\delta x)^2}\left(\phi_{\boldsymbol{x}+\delta x\hat{\boldsymbol{\mu}}_i} - \phi_{\boldsymbol{x}} - \left(\phi_{\boldsymbol{x}} - \phi_{\boldsymbol{x}-\delta x\hat{\boldsymbol{\mu}}_i}\right)\right) \end{aligned} \tag{79}$$

giving

$$\frac{\partial^2}{\partial x_i^2}\phi_{\boldsymbol{x}} \approx \frac{1}{(\delta x)^2}\left(-2\phi_{\boldsymbol{x}} + \phi_{\boldsymbol{x}+\delta x\hat{\boldsymbol{\mu}}_i} + \phi_{\boldsymbol{x}-\delta x\hat{\boldsymbol{\mu}}_i}\right). \tag{80}$$

Applying this approximation for each spatial dimension in (73) we can approximate the Laplacian operator acting in our discretized system:

$$\boldsymbol{\nabla}^2\phi(\boldsymbol{x}) \approx \left[\boldsymbol{\nabla}^2\phi\right]_{\boldsymbol{x}} = \sum_{i=1}^{d} \frac{1}{(\delta x)^2}\left[-2\phi_{\boldsymbol{x}} + \phi_{\boldsymbol{x}+\delta x\hat{\boldsymbol{\mu}}_i} + \phi_{\boldsymbol{x}-\delta x\hat{\boldsymbol{\mu}}_i}\right] \tag{81}$$

B. Twisted Boundary Conditions

We begin with $\phi_{\boldsymbol{x}+L\hat{\boldsymbol{\mu}}_i}$ where $L = N\delta x$ is the spatial lattice extent. Using (9) to go to the momentum-space lattice we can write

$$\begin{aligned}\phi_{\boldsymbol{x}+L\hat{\boldsymbol{\mu}}_i} &= \frac{1}{(2\pi)^d}\sum_{\boldsymbol{k}} e^{-i\boldsymbol{k}\cdot(\boldsymbol{x}+L\hat{\boldsymbol{\mu}}_i)}\phi_{\boldsymbol{k}} \\ &= \frac{1}{(2\pi)^d}\sum_{\boldsymbol{k}} e^{-i\boldsymbol{k}\cdot\boldsymbol{x}}\phi_{\boldsymbol{k}} e^{-i\boldsymbol{k}\cdot L\hat{\boldsymbol{\mu}}_i}.\end{aligned} \tag{82}$$

Using the definition of the momentum-space lattice (8) we can write

$$\begin{aligned}e^{-i\boldsymbol{k}\cdot L\hat{\boldsymbol{\mu}}_i} &= e^{-i(-\boldsymbol{k}_{max}+(\boldsymbol{s}-\boldsymbol{\Delta})\delta k)\cdot L\hat{\boldsymbol{\mu}}_i} \\ &= e^{-i(-n_0\delta kL+(\boldsymbol{s}_i-\boldsymbol{\Delta}_i)\delta kL)},\end{aligned} \tag{83}$$

and now using the fact that $\delta k = \frac{2\pi}{L}$

$$\begin{aligned}&= e^{i(n_0-\boldsymbol{s}_i)2\pi}e^{i2\pi\boldsymbol{\Delta}_i} \\ &= e^{i2\pi\boldsymbol{\Delta}_i}\end{aligned} \tag{84}$$

since $n_0, \boldsymbol{s}_i \in \mathbb{Z}$. We hence have

$$\phi_{\boldsymbol{x}+L\hat{\boldsymbol{\mu}}_i} = \frac{1}{(2\pi)^d}\sum_{\boldsymbol{k}} e^{-i\boldsymbol{k}\cdot\boldsymbol{x}}\phi_{\boldsymbol{k}} e^{i2\pi\boldsymbol{\Delta}_i} \tag{85}$$

which simplifies to

$$\phi_{\boldsymbol{x}+L\hat{\boldsymbol{\mu}}_{\boldsymbol{i}}} = e^{i(2\pi\Delta_i)}\phi_{\boldsymbol{x}} \tag{86}$$

by using (9) to go back to position space. Thus, half-integer values of $\boldsymbol{\Delta}_i$ amount to a relative minus sign for each wrap around the lattice, corresponding to twisted B.C.s.

C. Fourier Transform of the Discretized, Free Hamiltonian

Defined on the spatial lattice, the discretized Hamiltonian for free scalar field theory takes the form

$$\hat{H} = \frac{1}{2}\sum_{\boldsymbol{x}}\left[\hat{\Pi}_{\boldsymbol{x}}^2 - \hat{\phi}_{\boldsymbol{x}}\left[\hat{\boldsymbol{\nabla}}^2\hat{\phi}\right]_{\boldsymbol{x}} + {m_0}^2\hat{\phi}_{\boldsymbol{x}}^2\right]. \tag{87}$$

Writing explicitly the finite-difference definition of the discretized Laplacian (Appendix A):

$$\hat{H} = \frac{1}{2}\sum_{\boldsymbol{x}}\left[\hat{\Pi}_{\boldsymbol{x}}^2 + \sum_{i=1}^{d}\frac{1}{(\delta x)^2}[2\hat{\phi}_{\boldsymbol{x}}^2 - \hat{\phi}_{\boldsymbol{x}}\hat{\phi}_{\boldsymbol{x}+\delta x\hat{\boldsymbol{\mu}}_i} - \hat{\phi}_{\boldsymbol{x}}\hat{\phi}_{\boldsymbol{x}-\delta x\hat{\boldsymbol{\mu}}_i}] + {m_0}^2\hat{\phi}_{\boldsymbol{x}}^2\right]. \tag{88}$$

Performing a Fourier transform (9) to momentum-space yields

$$\hat{H} = \frac{1}{2} \cdot \frac{1}{(2\pi)^{2d}} \sum_{\boldsymbol{x}} \sum_{\boldsymbol{k},\boldsymbol{k'}} \Big[e^{-i\boldsymbol{k}\cdot\boldsymbol{x}} \hat{\Pi}_{\boldsymbol{k}} e^{-i\boldsymbol{k'}\cdot\boldsymbol{x}} \hat{\Pi}_{\boldsymbol{k'}}$$

$$+ \frac{1}{(\delta x)^2} \sum_{i=1}^{d} \Big[2e^{-i\boldsymbol{k}\cdot\boldsymbol{x}} \hat{\phi}_{\boldsymbol{k}} e^{-i\boldsymbol{k'}\cdot\boldsymbol{x}} \hat{\phi}_{\boldsymbol{k'}} - e^{-i\boldsymbol{k}\cdot\boldsymbol{x}} \hat{\phi}_{\boldsymbol{k}}\, e^{-i\boldsymbol{k'}\cdot(\boldsymbol{x}-\delta x \hat{\boldsymbol{\mu}}_i)} \hat{\phi}_{\boldsymbol{k'}} - e^{-i\boldsymbol{k}\cdot\boldsymbol{x}} \hat{\phi}_{\boldsymbol{k}} e^{-i\boldsymbol{k'}\cdot(\boldsymbol{x}+\delta x \hat{\boldsymbol{\mu}}_i)} \hat{\phi}_{\boldsymbol{k'}} \Big]$$

$$+ m_0^2 e^{-i\boldsymbol{k}\cdot\boldsymbol{x}}\, \hat{\phi}_{\boldsymbol{k}} e^{-i\boldsymbol{k'}\cdot\boldsymbol{x}} \hat{\phi}_{\boldsymbol{k'}} \Big] \tag{89}$$

$$= \frac{1}{2} \cdot \frac{1}{(2\pi)^{2d}} \sum_{\boldsymbol{k},\boldsymbol{k'}} \sum_{\boldsymbol{x}} e^{i\boldsymbol{x}\cdot(-\boldsymbol{k}-\boldsymbol{k'})} \left[\hat{\Pi}_{\boldsymbol{k}} \hat{\Pi}_{\boldsymbol{k'}} + \left[m_0^2 + \frac{1}{(\delta x)^2} \sum_{i=1}^{d} \left[2 - e^{ik'_i \delta x} - e^{-ik'_i \delta x} \right] \right] \hat{\phi}_{\boldsymbol{k}} \hat{\phi}_{\boldsymbol{k'}} \right]. \tag{90}$$

Performing a change of variables $\boldsymbol{k} \to -\boldsymbol{k}$ and using the completeness relation (10)

$$= \frac{1}{2} \cdot \frac{1}{(2\pi)^{2d}} \sum_{\boldsymbol{k},\boldsymbol{k'}} \delta_{\boldsymbol{k},\boldsymbol{k'}} \left[\hat{\Pi}_{-\boldsymbol{k}} \hat{\Pi}_{\boldsymbol{k'}} + \left[m_0^2 + \frac{1}{(\delta x)^2} \sum_{i=1}^{d} \left[2 - e^{ik'_i \delta x} - e^{-ik'_i \delta x} \right] \right] \hat{\phi}_{-\boldsymbol{k}} \hat{\phi}_{\boldsymbol{k'}} \right] \tag{91}$$

$$= \frac{1}{2} \cdot \frac{1}{(2\pi)^{d}} \sum_{\boldsymbol{k}} \left[\hat{\Pi}_{-\boldsymbol{k}} \hat{\Pi}_{\boldsymbol{k}} + \left[m_0^2 + \frac{1}{(\delta x)^2} \sum_{i=1}^{d} \left[2 - e^{ik_i \delta x} - e^{-ik_i \delta x} \right] \right] \hat{\phi}_{-\boldsymbol{k}} \hat{\phi}_{\boldsymbol{k}} \right] \tag{92}$$

$$= \frac{1}{2} \cdot \frac{1}{(2\pi)^{d}} \sum_{\boldsymbol{k}} \left[\hat{\Pi}_{-\boldsymbol{k}} \hat{\Pi}_{\boldsymbol{k}} + \omega_{\boldsymbol{k}}^2 \hat{\phi}_{-\boldsymbol{k}} \hat{\phi}_{\boldsymbol{k}} \right] \tag{93}$$

where the frequency

$$\begin{aligned} \omega_{\boldsymbol{k}}^2 &= m_0^2 + \frac{1}{(\delta x)^2} \sum_{i=1}^{d} \left[2 - e^{ik_i \delta x} - e^{-ik_i \delta x} \right] \\ &= m_0^2 + \frac{4}{(\delta x)^2} \sum_{i=1}^{d} \left[\frac{e^{ik_i \delta x/2} - e^{-ik_i \delta x/2}}{2i} \right]^2 \\ &= m_0^2 + \frac{4}{(\delta x)^2} \sum_{i=1}^{d} \sin^2 \left(\frac{k_i \delta x}{2} \right) \end{aligned} \tag{94}$$

Hence, the free theory amounts to a collection of uncoupled harmonic oscillators on the momentum-space lattice.

C. Fourier Transform of $\tilde{\Pi}^2$

Defining our discrete Fourier transform analogously to (9):

$$\left| \tilde{\phi}^{\alpha} \right\rangle = \frac{1}{2\pi} \sum_{k_{\tilde{\phi}}} e^{-i\tilde{\phi}^{\alpha} k_{\tilde{\phi}}} \left| k_{\tilde{\phi}} \right\rangle \tag{95}$$

$\tilde{\Pi}^2$ acts in our digitized field space basis as

$$\tilde{\Pi}^2 \left| \tilde{\phi}^{\alpha} \right\rangle = \frac{1}{\left(\delta_{\tilde{\phi}}\right)^2} \left[2 \left| \tilde{\phi}^{\alpha} \right\rangle - \left| \tilde{\phi}^{\alpha+1} \right\rangle - \left| \tilde{\phi}^{\alpha-1} \right\rangle \right]. \tag{96}$$

By performing a Fourier transform this can be expressed in the conjugate-momentum basis:

$$\begin{aligned}
\tilde{\Pi}^2 \left|\tilde{\phi}^\alpha\right\rangle &= \frac{1}{2\pi}\sum_{k_{\tilde{\phi}}} \frac{1}{\left(\delta_{\tilde{\phi}}\right)^2}\left[2e^{-i\tilde{\phi}^\alpha k_{\tilde{\phi}}} - e^{-i\tilde{\phi}^{\alpha+1}k_{\tilde{\phi}}} - e^{-i\tilde{\phi}^{\alpha-1}k_{\tilde{\phi}}}\right]\left|k_{\tilde{\phi}}\right\rangle \\
&= \frac{1}{2\pi}\sum_{k_{\tilde{\phi}}} \frac{1}{\left(\delta_{\tilde{\phi}}\right)^2}\left[2e^{-i\tilde{\phi}^\alpha k_{\tilde{\phi}}} - e^{-i\left(\tilde{\phi}^\alpha+\delta_{\tilde{\phi}}\right)k_{\tilde{\phi}}} - e^{-i\left(\tilde{\phi}^\alpha-\delta_{\tilde{\phi}}\right)k_{\tilde{\phi}}}\right]\left|k_{\tilde{\phi}}\right\rangle \\
&= \frac{1}{2\pi}\sum_{k_{\tilde{\phi}}} e^{-i\tilde{\phi}^\alpha k_{\tilde{\phi}}}\frac{1}{\left(\delta_{\tilde{\phi}}\right)^2}\left[2 - e^{-i\delta_{\tilde{\phi}}k_{\tilde{\phi}}} - e^{i\delta_{\tilde{\phi}}k_{\tilde{\phi}}}\right]\left|k_{\tilde{\phi}}\right\rangle \\
&= \frac{1}{2\pi}\sum_{k_{\tilde{\phi}}} e^{-i\tilde{\phi}^\alpha k_{\tilde{\phi}}}\hat{k}_{\tilde{\phi}}\left|k_{\tilde{\phi}}\right\rangle
\end{aligned} \tag{97}$$

where

$$\begin{aligned}
\hat{k}_{\tilde{\phi}} &= \frac{1}{\left(\delta_{\tilde{\phi}}\right)^2}\left[2 - e^{-i\delta_{\tilde{\phi}}k_{\tilde{\phi}}} - e^{i\delta_{\tilde{\phi}}k_{\tilde{\phi}}}\right] \\
&= \frac{4}{\left(\delta_{\tilde{\phi}}\right)^2}\left[\frac{e^{i\delta_{\tilde{\phi}}k_{\tilde{\phi}}/2} - e^{-i\delta_{\tilde{\phi}}k_{\tilde{\phi}}/2}}{2i}\right]^2 \\
&= \frac{4}{\left(\delta_{\tilde{\phi}}\right)^2}\sin^2\left(\frac{\delta_{\tilde{\phi}}k_{\tilde{\phi}}}{2}\right).
\end{aligned} \tag{98}$$

We could also Fourier transform the initial state $\left|\tilde{\phi}^\alpha\right\rangle$ to express:

$$\begin{aligned}
\tilde{\Pi}^2 \left|\tilde{\phi}^\alpha\right\rangle &= \tilde{\Pi}^2\frac{1}{2\pi}\sum_{k_{\tilde{\phi}}} e^{-i\tilde{\phi}^\alpha k_{\tilde{\phi}}}\left|k_{\tilde{\phi}}\right\rangle \\
&= \frac{1}{2\pi}\sum_{k_{\tilde{\phi}}} e^{-i\tilde{\phi}^\alpha k_{\tilde{\phi}}}\tilde{\Pi}^2\left|k_{\tilde{\phi}}\right\rangle
\end{aligned} \tag{99}$$

Finally, by comparing (97) and (99) we arrive at the result

$$\tilde{\Pi}^2\left|k_{\tilde{\phi}}\right\rangle = \hat{k}_{\tilde{\phi}}\left|k_{\tilde{\phi}}\right\rangle, \tag{100}$$

with

$$\hat{k}_{\tilde{\phi}} = \frac{4}{\left(\delta_{\tilde{\phi}}\right)^2}\sin^2\left(\frac{\delta_{\tilde{\phi}}k_{\tilde{\phi}}}{2}\right) \tag{101}$$

in agreement with (27).

D. Decomposition of $\tilde{\phi}^2$ and $\bar{\Pi}^2$ for $n_q = 3$ Qubits.

It was explained in Section 3.2.*iii* that upon identifying our digitized basis with the computational basis of our qubit space, the action of an operator in this basis can be decomposed into tensor products of single qubit unitary operators. This process is carried out one coefficient at a time and is outlined in the following example for the decomposition of $\tilde{\phi}^2$ and $\bar{\Pi}^2$ under a $n_q = 3$ qubit JLP digitization..

We define the general tensor product of 3 single-qubit unitary gates as

$$T^{i,j,k} = \bar{\sigma}^i \otimes \bar{\sigma}^j \otimes \bar{\sigma}^k \quad \text{where} \quad \bar{\sigma} = \left(\sigma^x, \sigma^y, \sigma^z, \hat{\mathbb{I}}_2\right) \tag{102}$$

i.e. $i, j, k \in \{0, 1, 2, 3\}$. Beginning with $\tilde{\phi}^2$, our operator can be expressed as a sum of these general tensor products

$$\tilde{\phi}^2 = \sum_{ijk} C_{ijk} T^{ijk} \tag{103}$$

with coefficients given by (34) such that

$$C_{i'j'k'} = \frac{1}{8} Tr\left[\tilde{\phi}^2 T^{i'j'k'}\right] = \frac{1}{8} Tr\left[T^{i'j'k'} \tilde{\phi}^2\right]. \tag{104}$$

Writing our operator $\tilde{\phi}^2$ as a projector in the computational basis

$$\begin{aligned} \tilde{\phi}^2 = &\left(\tilde{\phi}^0\right)^2 |000\rangle\langle 000| + \left(\tilde{\phi}^1\right)^2 |001\rangle\langle 001| + \left(\tilde{\phi}^2\right)^2 |010\rangle\langle 010| + \left(\tilde{\phi}^3\right)^2 |011\rangle\langle 011| \\ &+ \left(\tilde{\phi}^4\right)^2 |100\rangle\langle 100| + \left(\tilde{\phi}^5\right)^2 |101\rangle\langle 101| + \left(\tilde{\phi}^6\right)^2 |110\rangle\langle 110| + \left(\tilde{\phi}^7\right)^2 |111\rangle\langle 111| \end{aligned} \tag{105}$$

it is straightforward to evaluate the action of a specific product of single-qubit unitaries, $T^{i'j'k'} = \bar{\sigma}^{i'} \otimes \bar{\sigma}^{j'} \otimes \bar{\sigma}^{k'}$, on our operator before taking the trace to determine the corresponding coefficient $C_{i'j'k'}$. We can simplify the present case by realising that since ϕ is diagonal, any tensor products including Pauli x or Pauli y gates will not contribute to our decomposition, since these operations induce bit flips when acting on qubits in the computational basis and hence result in a vanishing trace. As such, we only consider coefficients C_{ijk} s.t. $i, j, k \in \{2, 3\}$. For example, we have

$$\begin{aligned} C_{233} &= \frac{1}{8} Tr\left[\left(\left(\hat{\mathbb{I}}_2 \otimes \sigma^z \otimes \sigma^z\right) \tilde{\phi}^2\right]\right. \\ &= \frac{1}{8}\left[\left(\tilde{\phi}^0\right)^2 - \left(\tilde{\phi}^1\right)^2 - \left(\tilde{\phi}^2\right)^2 + \left(\tilde{\phi}^3\right)^2 + \left(\tilde{\phi}^4\right)^2 - \left(\tilde{\phi}^5\right)^2 - \left(\tilde{\phi}^6\right)^2 + \left(\tilde{\phi}^7\right)^2\right] \\ &= \frac{\tilde{\phi}^2_{max}}{49} \frac{1}{8}\left[(-7)^2 - (-5)^2 - (-3)^2 + (-1)^2 + 1^2 - 3^2 - 5^2 + 7^2\right] \\ &= \frac{4}{49} \tilde{\phi}^2_{max} \end{aligned} \tag{106}$$

where in the second equality we used the action of σ^z in the computational basis: $\sigma^z |0\rangle = +|0\rangle$, $\sigma^z |1\rangle = -|1\rangle$ and in the third equality we used (21) to write our sample field values in terms of $\bar{\phi}_{max}$. Repeating for the remaining coefficients and analogously for the $\bar{\Pi}^2$ operator[18] we find that

$$\tilde{\phi}^2 = \frac{4}{49} \tilde{\phi}^2_{max} \mathcal{O}_0^{(n_q=3)}, \quad \bar{\Pi}^2 = \frac{49\pi^2}{64 \bar{\phi}^2_{max}} \mathcal{O}_0^{(n_q=3)} \tag{107}$$

where

$$\mathcal{O}_0^{(n_q=3)} = 4\sigma^z \otimes \sigma^z \otimes \hat{\mathbb{I}}_2 + 2\sigma^z \otimes \hat{\mathbb{I}}_2 \otimes \sigma^z + \hat{\mathbb{I}}_2 \otimes \sigma^z \otimes \sigma^z + \frac{21}{4} \hat{\mathbb{I}}_8 \tag{108}$$

[18] In this case, the computational basis states spanning our qubit space are identified with the *conjugate-momentum* basis in which the $\bar{\Pi}^2$ operator is diagonal

s.t. they share the same operator structure when acting in their respective bases, in agreement with the findings of KS [22]. This is a consequence of our digitized field space and conjugate-momentum space bases sharing the same structure by construction.

E. Deriving the δH_{ω_ϕ} and H_{int} Matrices

Here we present the derivations of (48) and (51), before using them to derive the matrices representing the raw truncation of δH_{ω_ϕ} and H_{int} to our HO digitization basis.

We begin by expressing $\tilde{\phi}$ in terms of creation/annihilation operators of our basis HO:

$$\bar{\phi} = \frac{1}{\sqrt{2\omega_\phi}} \left(a^\dagger_{\omega_\phi} + a_{\omega_\phi} \right). \tag{109}$$

The creation/annihilation operators act on the HO eigenstates states as

$$.a^\dagger_{\omega_\phi} |n\rangle = \sqrt{n+1}\, |n+1\rangle\,, \quad a_{\omega_\phi} |n\rangle = \sqrt{n}\, |n-1\rangle \tag{110}$$

To derive the action of $\tilde{\phi}^2$ on the HO eigenstates (48) we express it in terms of creation/annihilation operators and evaluate their action on the HO eigenstates using (110):

$$\begin{aligned} \bar{\phi}^2 |n\rangle &= \frac{1}{2\omega_\phi} \left(a^\dagger_{\omega_\phi} + a_{\omega_\phi} \right)^2 |n\rangle \\ &= \frac{1}{2\omega_\phi} \left(a^\dagger_{\omega_\phi} + a_{\omega_\phi} \right) \left[\sqrt{n+1}\, |n+1\rangle + \sqrt{n}\, |n-1\rangle \right] \\ &= \frac{1}{2\omega_\phi} \left[\sqrt{(n+2)(n+1)}\, |n+2\rangle + (2n+1)\, |n\rangle + \sqrt{n(n-1)}\, |n-2\rangle \right]. \end{aligned} \tag{111}$$

The action of $\tilde{\phi}^4$ (51) is derived in the same way:

$$\begin{aligned} \tilde{\phi}^4 |n\rangle &= \frac{1}{4\omega_\phi^2} \left(a^\dagger_{\omega_\phi} + a_{\omega_\phi} \right)^4 |n\rangle \\ &= \cdots \\ &= \frac{1}{4\omega_\phi^2} \Big[\sqrt{(n+4)(n+3)(n+2)(n+1)}\, |n+4\rangle \\ &\quad + (4n+6) \sqrt{(n+2)(n+1)}\, |n+2\rangle + (6n^2+6n+3)\, |n\rangle \\ &\quad + (4n-2) \sqrt{n(n-1)}\, |n-2\rangle + \sqrt{n(n-1)(n-2)(n-3)}\, |n-4\rangle \Big]. \end{aligned} \tag{112}$$

Defining our digitization basis as the lowest n_s eigenstates of our basis HO, i.e. $\{|n\rangle\, | n < n_s\}$, we can use (48) and (51) to derive the matrix representations of δH_{ω_ϕ} and H_{int} acting in our digitized basis. We have

$$\begin{aligned} \left(\delta H_{\omega_\phi} \right)_{m,n} &= \langle m |\, \delta H_{\omega_\phi}\, |n\rangle \\ &= \frac{1}{2} \left(1 - \omega_\phi^2 \right) \langle m |\, \bar{\phi}^2\, |n\rangle \\ &= \frac{1}{2} \left(1 - \omega_\phi^2 \right) \frac{1}{2\omega_\phi} \Big[\sqrt{(n+2)(n+1)}\, \langle m | n+2 \rangle \\ &\quad + (2n+1)\, \langle m | n \rangle + \sqrt{n(n-1)}\, \langle m | n-2 \rangle \Big] \end{aligned} \tag{113}$$

where in the third equality we used (48). Now, by orthonormality of the $|m\rangle, |n\rangle$

$$\left(\delta H_{\omega_\phi}\right)_{m,n} = \frac{1}{2}\left(1-\omega_\phi^2\right)\frac{1}{2\omega_\phi}\left[\sqrt{(n+2)(n+1)}\delta_{m,n+2} + (2n+1)\delta_{m,n} + \sqrt{n(n-1)}\delta_{m,n-2}\right] \tag{114}$$

for $n, m < n_s$. We can now use (114) to define our matrix element by element. For example, with $n_q = 3$ ($n_s = 8$ basis states) we get

$$\delta H_{\omega_\phi} = \frac{\left(1-\omega_\phi^2\right)}{4\omega_\phi}\begin{bmatrix} 1 & 0 & \sqrt{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & \sqrt{6} & 0 & 0 & 0 & 0 \\ \sqrt{2} & 0 & 5 & 0 & 2\sqrt{3} & 0 & 0 & 0 \\ 0 & \sqrt{6} & 0 & 7 & 0 & 2\sqrt{5} & 0 & 0 \\ 0 & 0 & 2\sqrt{3} & 0 & 9 & 0 & \sqrt{30} & 0 \\ 0 & 0 & 0 & 2\sqrt{5} & 0 & 11 & 0 & \sqrt{42} \\ 0 & 0 & 0 & 0 & \sqrt{30} & 0 & 13 & 0 \\ 0 & 0 & 0 & 0 & 0 & \sqrt{42} & 0 & 15 \end{bmatrix}. \tag{115}$$

For H_{int} we analogously use (51) to obtain

$$\begin{aligned} (H_{int}) &= \frac{\bar{\lambda}_0}{4!}\langle m|\,\bar{\phi}^4\,|n\rangle \\ &= \cdots \\ &= \frac{\bar{\lambda}_0}{4!}\frac{1}{4\omega_\phi^2}\Big[\sqrt{(n+4)(n+3)(n+2)(n+1)}\delta_{m,n+4} \\ &\quad + (4n+6)\sqrt{(n+2)(n+1)}\delta_{m,n+2} + (6n^2+6n+3)\delta_{m,n} \\ &\quad + (4n-2)\sqrt{n(n-1)}\delta_{m,n-2} + \sqrt{n(n-1)(n-2)(n-3)}\delta_{m,n-4}\Big], \end{aligned} \tag{116}$$

which we can similarly use to derive H_{int} element by element. Continuing with the $n_q = 3$ example, we get

$$H_{int=} = \frac{\bar{\lambda}_0}{4!}\frac{1}{4\omega_\phi^2}\begin{bmatrix} 3 & 0 & 6\sqrt{2} & 0 & 2\sqrt{6} & 0 & 0 & 0 \\ 0 & 15 & 0 & 10\sqrt{6} & 0 & 2\sqrt{30} & 0 & 0 \\ 6\sqrt{2} & 0 & 39 & 0 & 28\sqrt{3} & 0 & 6\sqrt{10} & 0 \\ 0 & 10\sqrt{6} & 0 & 75 & 0 & 36\sqrt{5} & 0 & 2\sqrt{210} \\ 2\sqrt{6} & 0 & 28\sqrt{3} & 0 & 123 & 0 & 22\sqrt{30} & 0 \\ 0 & 2\sqrt{30} & 0 & 36\sqrt{5} & 0 & 183 & 0 & 26\sqrt{42} \\ 0 & 0 & 6\sqrt{10} & 0 & 22\sqrt{30} & 0 & 255 & 0 \\ 0 & 0 & 0 & 2\sqrt{210} & 0 & 26\sqrt{42} & 0 & 339 \end{bmatrix}. \tag{117}$$

F. HO basis Equivalent of $\bar{\phi}_{max}$

In the JLP formalism, $\bar{\phi}_{max}$ is a measure of field-space truncation, corresponding to half the range of the truncated field space. In the HO formalism we have no single variable to directly tune field-space coverage; instead we have that both n_q and ω_ϕ jointly determine the coverage

of field space attained by the digitization. For the purposes of comparing the two approaches to digitization in FIG.7 we must define a $\bar{\phi}_{max}$ equivalent for the HO regime.

The half-width at half-maximum (HWHM) of a gaussian is defined $\sqrt{2\ln 2} \times \sigma \approx 1.4\sigma$, where $\sigma = \sqrt{\langle \bar{\phi}^2 \rangle}$ is the standard deviation. We can generalise this notion to the Hermite-Gauss functions of order n that correspond to our HO basis states, to define the 'half-width' in field space of the nth basis state $|n\rangle$ as $1.4\sqrt{\langle \bar{\phi}^2 \rangle_n}$. We have

$$\langle \bar{\phi}^2 \rangle_n = \langle n | \bar{\phi}^2 | n \rangle = \frac{2n+1}{2\omega_\phi} \tag{118}$$

by (48), such that the 'half-width' of the nth basis state in field space is $1.4\sqrt{\frac{n+1/2}{\omega_\phi}}$. We now define our $\bar{\phi}_{max}$ equivalent variable as the 'half-width' in field space of the 'widest' basis state, corresponding to $n = n_s - 1 = 2^{n_q} - 1$. We hence have

$$\bar{\phi}_{max} \simeq 1.4\sqrt{\frac{2^{n_q} - 1 + 1/2}{\omega_\phi}} \approx 1.4\sqrt{\frac{2^{n_q}}{\omega_\phi}} \tag{119}$$

G. Perturbative expansion of Σ and T

This appendix shows how to derive the perturbative expansions of Σ and T, following the methodology of CFHL [21] whilst filling in the gaps for completeness. The derivation begins with the interaction picture time evolution operator defined

$$U_{IP}(t_f, t_i) = Texp\left\{-i\int_{t_i}^{t_f} dt V_{IP}(t)\right\} \tag{120}$$

where our interaction picture interaction term is

$$V_{IP} = e^{iH_0 t} V e^{-\epsilon t} e^{-iH_0 t}. \tag{121}$$

$U_{IP}(t_f, t_i)$ hence satisfies the differential equation

$$\frac{\partial}{\partial t_i} U_{IP}(t_f, t_i) = i U_{IP}(t_f, t_i) V_{IP}(t_i) \tag{122}$$

with boundary condition $U_{IP}(t_f, t_f) = \mathbb{I}$, which solves to give

$$U_{IP}(t_f, t_i) = \mathbb{I} - i\int_{t_i}^{t_f} dt U_{IP}(t_f, t) V_{IP}(t). \tag{123}$$

The perturbative expansion of Σ is now derived by expanding (123) iteratively within the defi-

nition of $\langle f|\,\Sigma\,|i\rangle$. Iterating up to second order we get

$$\begin{aligned}
&\langle f|\,\Sigma(\epsilon)\,|i\rangle \equiv \lim_{t_f\to\infty}\langle f|\,U_{IP}\,(t_f,0)\,|i\rangle \\
&= \lim_{t_f\to\infty}\langle f|\left\{\mathbb{I} - i\int_0^{t_f} dt U_{IP}\,(t_f,t)\,V_{IP}(t)\right\}|i\rangle \\
&= \lim_{t_f\to\infty}\left\{\langle f|\,\mathbb{I}\,|i\rangle - i\,\langle f|\int_0^{t_f} dt\left\{\mathbb{I} - i\int_t^{t_f} dt' U_{IP}\,(t_f,t')\,V_{IP}(t')\right\}V_{IP}(t)\,|i\rangle\right\} \\
&= \delta_{fi} + \lim_{t_f\to\infty}\left\{-i\,\langle f|\int_0^{t_f} dt V_{IP}(t)\,|i\rangle\right. \\
&\quad \left. - \langle f|\int_0^{t_f} dt\int_t^{t_f} dt'\left\{\mathbb{I} - i\int_{t'}^{t_f} dt'' U_{IP}\,(t_f,t'')\,V_{IP}(t'')\right\}V_{IP}\,(t')\,V_{IP}\,(t)\,|i\rangle\right\} \\
&= \delta_{fi} + \lim_{t_f\to\infty}\left\{-i\,\langle f|\int_0^{t_f} dt V_{IP}(t)\,|i\rangle - \langle f|\int_0^{t_f} dt\int_t^{t_f} dt' V_{IP}\,(t')\,V_{IP}\,(t)\,|i\rangle\right\} + \mathcal{O}\left(V^3\right). \\
&= \delta_{fi} + \lim_{t_f\to\infty}\left\{-i\int_0^{t_f} dt\,\langle f|\,e^{iH_0t}Ve^{-\epsilon t}e^{-iH_0t}\,|i\rangle\right. \\
&\quad \left. - \int_0^{t_f} dt\int_t^{t_f} dt'\,\langle f|\,e^{iH_0t'}Ve^{-\epsilon t'}e^{-iH_0t'}e^{iH_0t}Ve^{-\epsilon t}e^{-iH_0t}\,|i\rangle\right\} + \mathcal{O}\left(V^3\right)
\end{aligned} \tag{124}$$

Evaluating the action of H_0 to the left and to the right and inserting the identity $\mathbb{I} = \sum_\alpha |\alpha\rangle\langle\alpha|$,

$$\begin{aligned}
&= \delta_{fi} + \lim_{t_f\to\infty}\left\{-i\,\langle f|\,V\,|i\rangle\int_0^{t_f} dt e^{i\left(E_f - E_i + i\epsilon\right)}\right. \\
&\quad \left. - \sum_\alpha\int_0^{t_f} dt\int_t^{t_f} dt' e^{iE_ft'}\,\langle f|\,Ve^{-\epsilon t'}e^{-H_0t'}\,|\alpha\rangle\,\langle\alpha|\,e^{iH_0t}Ve^{-\epsilon t}e^{-iE_it|i\rangle}\right\} + \mathcal{O}\left(V^3\right) \\
&= \delta_{fi} + \frac{\langle f|\,V\,|i\rangle}{E_{fi} + i\epsilon} \\
&\quad - \sum_\alpha\langle f|\,V\,|\alpha\rangle\,\langle\alpha|\,V\,|i\rangle\underbrace{\lim_{t_f\to\infty}\left\{\int_0^{t_f} dt\int_t^{t_f} dt' e^{i\left(E_{f\alpha} + i\epsilon\right)t'}e^{i(E_{\alpha i} + i\epsilon)t}\right\}}_{\circledast} + \mathcal{O}\left(V^3\right)
\end{aligned} \tag{125}$$

To evaluate $\circledast$, define $A = E_{f\alpha} + i\epsilon, \quad B = E_{\alpha i} + i\epsilon$. Then

$$\begin{aligned}
\circledast &= \lim_{t_f\to\infty}\left\{\int_0^{t_f} dt\int_t^{t_f} dt' e^{iAt'}e^{iBt}\right\} \\
&= \lim_{t_f\to\infty}\frac{1}{iA}\int_0^{t_f} dt\left\{e^{iAt_f}e^{iBt} - e^{iAt}e^{iBt}\right\} \\
&= \lim_{t_f\to\infty}\frac{1}{iA}\left\{\frac{1}{iB}e^{At_f}\left(e^{iBt_f} - 1\right) - \frac{1}{i\left(A + B\right)}\left(e^{i(A+B)t_f} - 1\right)\right\} \\
&= \frac{1}{iA}\cdot\frac{1}{i(A+B)} \\
&= -\frac{1}{\left(E_{f\alpha} + i\epsilon\right)\left(E_{fi} + 2i\epsilon\right)} \\
&= -\frac{1}{\left(E_{f\alpha} + i\epsilon\right)\left(E_{fi} + i\epsilon\right)}
\end{aligned} \tag{126}$$

where in the last line we used the fact that ϵ is small. Now,

$$\langle f|\,\Sigma(\epsilon)\,|i\rangle = \delta_{fi} + \frac{\langle f|\,V\,|i\rangle}{E_{fi}+i\epsilon} - \sum_{\alpha}\frac{\langle f|\,V\,|\alpha\rangle\,\langle\alpha|\,V\,|i\rangle}{(E_{f\alpha}+i\epsilon)\,(E_{fi}+i\epsilon)} + \mathcal{O}\left(V^3\right). \tag{127}$$

For convenience, we now remove the $\frac{1}{E_{fi}}$ by defining

$$\langle f|\Sigma|i\rangle = \delta_{fi} + \frac{\langle f|T|i\rangle}{E_{fi}+i\epsilon} \tag{128}$$

such that

$$\langle f|T|i\rangle = \langle f|V|i\rangle + \sum_{\alpha}\frac{\langle f|V|\alpha\rangle\,\langle\alpha|V|i\rangle}{(E_{f\alpha}+i\epsilon)} + \mathcal{O}\left(V^3\right). \tag{129}$$

H. Matching the Transition Matrix

The matching conditions given in [21], presented here in (64), are derived by matching the transition matrix defined above between the effective theory and the full theory. To do this we must first define the transition matrix in the effective theory, T_{eff}. Following [21], we define the effective interaction term

$$V_{eff} = H_1 + H_2 + H_3 + \mathcal{O}(V^4) \quad \text{s.t.} \quad H_{eff} = H_0 + V_{eff}. \tag{130}$$

The perturbative expansion of T_{eff} is then defined by restricting the perturbative expansion of T (63) to $\mathcal{H}_{eff}$ and setting $V \to V_{eff}$.

$$\begin{aligned}\langle f|T_{eff}|i\rangle_{eff} = \langle f|V_{eff}|i\rangle &+ \sum_{\alpha}^{<}\frac{\langle f|V_{eff}|\alpha\rangle\,\langle\alpha|V_{eff}|i\rangle}{E_{f\alpha}} \\ &+ \sum_{\alpha,\beta}^{<}\frac{\langle f|V_{eff}|\alpha\rangle\,\langle\alpha|V_{eff}|\beta\rangle\,\langle\beta|V_{eff}|i\rangle}{E_{f\alpha}E_{f\beta}} + \mathcal{O}(V_{eff}^4)\end{aligned} \tag{131}$$

where $\sum_{\alpha}^{<} = \sum_{E_\alpha<E_T}$

To derive the matching conditions we expand V_{eff} (130) within the expansion for T_{eff} and match with the expansion for T order by order. Here we will demonstrate the matching up to 3rd order. We have

$$\begin{aligned}\langle f|T|i\rangle &= \langle f|V|i\rangle + \sum_{\alpha}\frac{\langle f|V|\alpha\rangle\,\langle\alpha|V|i\rangle}{(E_{f\alpha}+i\epsilon)} + \sum_{\alpha,\beta}\frac{\langle f|V|\alpha\rangle\,\langle\alpha|V|\beta\rangle\,\langle\beta|V|i\rangle}{(E_{f\alpha}+i\epsilon)(E_{f\beta}+i\epsilon)} + \mathcal{O}(V^4) \\ &= \langle f|V|i\rangle + \Bigg(\underbrace{\sum_{\alpha}^{<}}_{I} + \sum_{\alpha}^{>}\Bigg)\frac{\langle f|V|\alpha\rangle\,\langle\alpha|V|i\rangle}{(E_{f\alpha}+i\epsilon)} \\ &\quad + \Bigg(\underbrace{\sum_{\alpha,\beta}^{<}}_{II} + \sum_{\alpha,\beta}^{>} + \sum_{\alpha}^{<}\sum_{\beta}^{>} + \underbrace{\sum_{\alpha}^{>}\sum_{\beta}^{<}}_{III}\Bigg)\frac{\langle f|V|\alpha\rangle\,\langle\alpha|V|\beta\rangle\,\langle\beta|V|i\rangle}{(E_{f\alpha}+i\epsilon)(E_{f\beta}+i\epsilon)} + \mathcal{O}(V^4)\end{aligned} \tag{132}$$

and

$$\begin{aligned}\langle f|T_{eff}|i\rangle_{eff} &= \langle f|\,H_1\,|i\rangle + \langle f|\,H_2\,|i\rangle + \langle f|\,H_3\,|i\rangle \\ &+ \underbrace{\sum_{\alpha}^{<} \frac{\langle f|H_1|\alpha\rangle\,\langle\alpha|H_1|i\rangle}{E_{f\alpha}}}_{i} + \sum_{\alpha}^{<} \frac{\langle f|H_1|\alpha\rangle\,\langle\alpha|H_2|i\rangle}{E_{f\alpha}} + \underbrace{\sum_{\alpha}^{<} \frac{\langle f|H_2|\alpha\rangle\,\langle\alpha|H_1|i\rangle}{E_{f\alpha}}}_{iii} \\ &+ \underbrace{\sum_{\alpha,\beta}^{<} \frac{\langle f|H_1|\alpha\rangle\,\langle\alpha|H_1|\beta\rangle\,\langle\beta|H_1|i\rangle}{E_{f\alpha}E_{f\beta}}}_{ii} + \mathcal{O}(V^4)\end{aligned} \tag{133}$$

Matching the first order terms trivially yields

$$\langle f|\,H_1\,|i\rangle_{eff} = \langle f|\,V\,|i\rangle\,. \tag{134}$$

Substituting this into (133) and matching 2nd order terms we find that i cancels I giving

$$\langle f|\,H_2\,|i\rangle_{eff} = \sum_{\alpha}^{>} \frac{\langle f|V|\alpha\rangle\,\langle\alpha|V|i\rangle}{E_{f\alpha}} \tag{135}$$

Again, substituting into (133) and matching 3rd order terms we find that ii cancels II and iii cancels III after a relabelling, leaving us with

$$\begin{aligned}&\langle f|\,H_3\,|i\rangle_{eff} + \sum_{\alpha}^{<}\sum_{\beta}^{>} \frac{\langle f|\,V\,|\alpha\rangle\,\langle\alpha|\,V\,|\beta\rangle\,\langle\beta|\,V\,|i\rangle}{E_{f\alpha}E_{\alpha\beta}} \\ &\qquad = \left(\sum_{\alpha,\beta}^{>} + \sum_{\alpha}^{<}\sum_{\beta}^{>}\right) \frac{\langle f|\,V\,|\alpha\rangle\,\langle\alpha|\,V\,|\beta\rangle\,\langle\beta|\,V\,|i\rangle}{E_{f\alpha}E_{f\beta}} \\ \implies\ &\langle f|\,H_3\,|i\rangle_{eff} = \sum_{\alpha,\beta}^{>} \frac{\langle f|\,V\,|\alpha\rangle\,\langle\alpha|\,V\,|\beta\rangle\,\langle\beta|\,V\,|i\rangle}{E_{f\alpha}E_{f\beta}} \\ &\qquad + \sum_{\alpha}^{<}\sum_{\beta}^{>} \frac{\langle f|\,V\,|\alpha\rangle\,\langle\alpha|\,V\,|\beta\rangle\,\langle\beta|\,V\,|i\rangle}{E_{f\alpha}E_{f\beta}E_{\alpha\beta}} \underbrace{(E_{\alpha\beta} - E_{f\beta})}_{=-E_{f\alpha}}\end{aligned} \tag{136}$$

and hence

$$\langle f|\,H_3\,|i\rangle_{eff} = \sum_{\alpha,\beta}^{>} \frac{\langle f|\,V\,|\alpha\rangle\,\langle\alpha|\,V\,|\beta\rangle\,\langle\beta|\,V\,|i\rangle}{E_{f\alpha}E_{f\beta}} - \sum_{\alpha}^{<}\sum_{\beta}^{>} \frac{\langle f|\,V\,|\alpha\rangle\,\langle\alpha|\,V\,|\beta\rangle\,\langle\beta|\,V\,|i\rangle}{E_{f\beta}E_{\alpha\beta}} \tag{137}$$

I. The H_2 Matrices

In Section 6.2 we found that after substituting the interaction term $V = \frac{1}{2}\left(1 - \omega_\phi^2\right)\bar{\phi}^2 + \frac{\bar{\lambda}_0}{4!}\bar{\phi}^4$ into the second order matching condition (64), the second order term in the effective Hamiltonian decomposes into four sub-terms such that

$$H_2 = H^{(2)}_{\bar{\phi}^2\bar{\phi}^2} + H^{(2)}_{\bar{\phi}^4\bar{\phi}^4} + H^{(2)}_{\bar{\phi}^2\bar{\phi}^4} + H^{(2)}_{\bar{\phi}^4\bar{\phi}^2}, \tag{138}$$

where

$$\begin{aligned}
\langle m|\, H^{(2)}_{\bar{\phi}^2\bar{\phi}^2}\, |n\rangle_{eff} &= \frac{1}{4}\left(1-\omega_\phi^2\right)^2 \sum_{\alpha}^{>} \frac{\langle m|\, \bar{\phi}^2\, |\alpha\rangle\, \langle\alpha|\, \bar{\phi}^2\, |n\rangle}{E_{m\alpha}} \\
\langle m|\, H^{(2)}_{\bar{\phi}^4\bar{\phi}^4}\, |n\rangle_{eff} &= \left(\frac{\bar{\lambda}_0}{4!}\right)^2 \sum_{\alpha}^{>} \frac{\langle m|\, \bar{\phi}^4\, |\alpha\rangle\, \langle\alpha|\, \bar{\phi}^4\, |n\rangle}{E_{m\alpha}} \\
\langle m|\, H^{(2)}_{\bar{\phi}^2\bar{\phi}^4}\, |n\rangle_{eff} &= \frac{1}{2}\left(1-\omega_\phi^2\right)\cdot\frac{\bar{\lambda}_0}{4!} \sum_{\alpha}^{>} \frac{\langle m|\, \bar{\phi}^2\, |\alpha\rangle\, \langle\alpha|\, \bar{\phi}^4\, |n\rangle}{E_{m\alpha}} \\
\langle m|\, H^{(2)}_{\bar{\phi}^4\bar{\phi}^2}\, |n\rangle_{eff} &= \frac{1}{2}\left(1-\omega_\phi^2\right)\cdot\frac{\bar{\lambda}_0}{4!} \sum_{\alpha}^{>} \frac{\langle m|\, \bar{\phi}^4\, |\alpha\rangle\, \langle\alpha|\, \bar{\phi}^2\, |n\rangle}{E_{m\alpha}}.
\end{aligned} \tag{139}$$

with $m, n < n_s$. Upon examination of formulae (48) and (51), we conclude that the infinite sums reduce to just a handful of terms corresponding to states just above the truncation energy. Expanding out the actions of $\bar{\phi}^2$, $\bar{\phi}^4$ via (48), (51) for the non-zero terms and evaluating the overlaps results in an expression of Kronecker deltas that can be used to define a matrix acting in the digitized basis. Explicitly, for the case of $H^{(2)}_{\bar{\phi}^2\bar{\phi}^4}$ (the other terms follow analogously) we get

$$\langle m|\, H^{(2)}_{\bar{\phi}^2\bar{\phi}^4}\, |n\rangle_{eff} = \frac{1}{2}\left(1-\omega_\phi^2\right)\cdot\frac{\bar{\lambda}_0}{4!}\left[\frac{\langle m|\, \bar{\phi}^2\, |n_s\rangle\, \langle n_s|\, \bar{\phi}^4\, |n\rangle}{\omega_\phi(m-n_s)} + \frac{\langle m|\, \bar{\phi}^2\, |n_s+1\rangle\, \langle n_s+1|\, \bar{\phi}^4\, |n\rangle}{\omega_\phi(m-n_s-1)}\right] \tag{140}$$

by inspection of (48). Now applying (48),(51) and expanding,

$$\begin{aligned}
&= \frac{1}{2}\left(1-\omega_\phi^2\right)\cdot\frac{\bar{\lambda}_0}{4!}\cdot\frac{1}{8\omega_\phi^4}\left[\frac{\sqrt{n_s(n_s-1)}}{m-n_s}\cdot(4n+6)\sqrt{(n+2)(n+1)}\,\langle m|n_s-2\rangle\,\langle n_s|n+2\rangle\right. \\
&\quad + \frac{\sqrt{n_s(n_s-1)}}{m-n_s}\cdot\sqrt{(n+4)(n+3)(n+2)(n+1)}\,\langle m|n_s-2\rangle\,\langle n_s|n+4\rangle \\
&\quad + \frac{\sqrt{(n_s+1)n_s}}{m-n_s-1}\cdot(4n+6)\sqrt{(n+2)(n+1)}\,\langle m|n_s-1\rangle\,\langle n_s+1|n+2\rangle \\
&\quad \left. + \frac{\sqrt{(n_s+1)n_s}}{m-n_s-1}\cdot\sqrt{(n+4)(n+3)(n+2)(n+1)}\,\langle m|n_s-1\rangle\,\langle n_s+1|n+4\rangle\right] \\
&= \frac{\left(1-\omega_\phi^2\right)}{16\omega_\phi^4}\cdot\frac{\bar{\lambda}_0}{4!}\Big[F(n_s)\delta_{m,n_s-2}\delta_{n,n_s-2} + G(n_s)\delta_{m,n_s-2}\delta_{n,n_s-4} \\
&\qquad\qquad + F(n_s+1)\delta_{m,n_s-1}\delta_{n,n_s-1} + G(n_s+1)\delta_{m,n_s-1}\delta_{n,n_s-3}\Big]
\end{aligned} \tag{141}$$

where

$$\begin{aligned}
F(n) &:= -\frac{n(n-1)(4n-2)}{2} \\
G(n) &:= -\frac{n(n-1)\sqrt{(n-2)(n-3)}}{2}
\end{aligned} \tag{142}$$

We can now use the Kronecker deltas to define the matrix. For n_q qubits such that $n_s = 2^{n_q}$ we get

$$\frac{\left(1-\omega_\phi^2\right)}{16\omega_\phi^4} \cdot \frac{\bar{\lambda}_0}{4!} \begin{bmatrix} 0 & \cdots & 0 & 0 & 0 & 0 \\ 0 & \cdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \cdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \cdots & G(n_s) & 0 & F(n_s) & 0 \\ 0 & \cdots & 0 & G(n_s+1) & 0 & F(n_s+1) \end{bmatrix}. \tag{143}$$

Repeating this process for the other terms we get that in the $n_q = 3$ ($n_s = 8$) case, the full second order term in the effective Hamiltonian is[19]

$$\begin{aligned} H_2^{n_q=3} =& \frac{\left(1-\omega_\phi^2\right)^2}{16\omega_\phi^3} \begin{bmatrix} 0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&A(8)&0 \\ 0&0&0&0&0&0&0&A(9) \end{bmatrix} \\ &+ \frac{1}{16\omega_\phi^5} \cdot \left(\frac{\bar{\lambda}_0}{4!}\right)^2 \begin{bmatrix} 0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0 \\ 0&0&0&0&E(8)&0&D(8)&0 \\ 0&0&0&0&0&E(9)&0&D(9) \\ 0&0&0&0&C(8)&0&B(8)+E(10)&0 \\ 0&0&0&0&0&C(9)&0&B(9)+E(11) \end{bmatrix} \\ &+ \frac{\left(1-\omega_\phi^2\right)}{16\omega_\phi^4} \cdot \frac{\bar{\lambda}_0}{4!} \begin{bmatrix} 0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&H(8)&0 \\ 0&0&0&0&0&0&0&H(9) \\ 0&0&0&0&G(8)&0&2F(8)&0 \\ 0&0&0&0&0&G(9)&0&2F(9) \end{bmatrix} \end{aligned} \tag{144}$$

[19] Note that $H^{(2)}_{\bar{\phi}^2\bar{\phi}^4}$ and $H^{(2)}_{\bar{\phi}^4\bar{\phi}^2}$ have been combined into one matrix.

for

$$\begin{aligned}
A(n) &:= -\frac{1}{2} \cdot n(n-1) \\
B(n) &:= -\frac{1}{2} \cdot n(n-1)(4n-2)^2 \\
C(n) &:= -\frac{1}{2} \cdot n(n-1)(4n-2)\sqrt{(n-2)(n-3)} \\
D(n) &:= -\frac{1}{4} \cdot n(n-1)(4n-2)\sqrt{(n-2)(n-3)} \\
E(n) &:= -\frac{1}{4} \cdot n(n-1)(n-2)(n-3) \\
F(n) &:= -\frac{1}{2} \cdot n(n-1)(4n-2) \\
G(n) &:= -\frac{1}{2} \cdot n(n-1)\sqrt{(n-2)(n-3)} \\
H(n) &:= -\frac{1}{4} \cdot n(n-1)\sqrt{(n-2)(n-3)}
\end{aligned} \quad . \tag{145}$$